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Analysis of aroma compounds in different grades of Luzhou Laojiao strong-aroma *Baijiu* by molecular sensory science

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ABSTRACT

Strong-aroma *Baijiu* is a beloved drink in China because of its elegant cellar aroma, sweetness and mellow taste, and its sales account for more than half of the *Baijiu* production. In this study, three different grades of strong-aroma *Baijiu* were analyzed systematically using comprehensive two-dimensional gas chromatography–time-of-flight mass spectrometry coupled with molecular sensory science. A total of 342 aroma compounds were identified in the three samples, including 152 aroma-active compounds. Principal component analysis revealed that higher-grade *Baijiu* contained more terpenes, acetals, and esters. In contrast, lower-grade *Baijiu* contained fewer esters and many compounds likely generated by the Maillard reaction, such as 5-methylfurfural, 2-propanoylfuran, 2,6-dimethylpyrazine, and 5-ethyl-2(5*H*)-furanone. Most of the aroma attributes of *Baijiu* samples could be simulated through aroma recombination experiments, but there were still some slight differences in individual aromas. Finally, the key aroma compounds of the three different grades of *Baijiu* were revealed through omission experiments. This study provided a scientific basis for the classification of strong-aroma *Baijiu* quality grades, thereby contributing to the high-quality development of the *Baijiu* industry.

1. Introduction

Baijiu is an alcoholic drink with traditional Chinese brewing characteristics (Fan et al., 2021). It has become one of the typical representatives of the world's distilled spirits because of its unique aroma (Zheng and Han, 2016a). Based on geographical location, production technology and environment, *Baijiu* can be classified into 12 categories, among which soy sauce aroma-type *Baijiu*, strong-aroma *Baijiu*, light aroma-type *Baijiu* and rice-type *Baijiu* are the four main aroma types (Wang et al., 2024). Strong-aroma *Baijiu* is favored by consumers in China because of its rich aroma and sweet taste, and its market share accounts for more than half of China's *Baijiu* production in recent years. (Wang et al., 2022). In particular, Luzhou Laojiao *Baijiu* brewed by traditional solid-state fermentation technology has elegant cellar aroma, soft taste and clean aftertaste, which makes it a typical representative of strong-aroma *Baijiu* (Li et al., 2012).

Aroma is one of the important indices for judging the quality and consumer choice of strong-aroma *Baijiu* (Duan et al., 2024). The aroma of strong-aroma *Baijiu* is caused mainly by the comprehensive action of trace compounds (Wang et al., 2022). At present, more than 800 volatile compounds have been identified in strong-aroma *Baijiu*, mainly including esters, aldehydes, alcohols, acids and other compounds (Wang et al., 2022). Zhao et al. (2018) identified 60 aroma compounds from strong-aroma *Baijiu* (Gujinggong *Baijiu*), among which ethyl hexanoate, ethyl octanoate, ethyl pentanoate, butyric acid and γ-nonalactone were identified as the key aroma compounds by aroma omission test. In particular, the flavor dilution (FD) value of ethyl hexanoate was as high as 59049. Mu et al. (2023) used HS-SPME combined with GC×GC–TOF MS to identify 168 volatile compounds from fresh strong-aroma *Baijiu*, among which 29 compounds (in particular ethyl hexanoate and ethyl lactate) were further identified as aroma-active compounds due to their high odor activity values (OAVs). Hong et al. (2023) identified 78 aroma compounds in strong-aroma *Baijiu* of Northern China by gas chromatography–olfactometry–mass spectrometry (GC–O–MS) combined with

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aroma extract dilution analysis (AEDA), OAV, aroma recombination and omission experiments. Among them, ethyl hexanoate, ethyl butanoate, hexanoic acid, butyric acid, ethyl acetate, ethyl pentanoate, and ethyl phenylpropanoate were demonstrated as the key aroma compounds.

The composition of aroma compounds in *Baijiu* is not only influenced by raw materials and fermentation, but also by the distillation process. Usually, the first distillate (except *Jiutou*) taken out during distillation process has high contents of esters and alcohols, and exhibits a high quality with a rich, pure, and harmonious aroma. The middle distillate obtained from the distillation contains moderate levels of esters, alcohols, acids, and aldehydes, showcasing relatively high quality. The later distillate (except *Jiuwei*) contains fewer and unbalanced aroma compounds, leading to astringent and inferior taste, and thereby lower quality (Ding et al., 2015). The different distillates are classified in quality by experienced *Baijiu* panelists, and then stored separately in containers, traditionally in ceramic jars, to serve as base *Baijiu* for later blending. Different grades of base *Baijiu* are stored for different periods of time. Typically, high-quality base *Baijiu* is stored for 2–3 years, medium-quality base *Baijiu* for 1–2 years, and low-quality base *Baijiu* for about half a year. *Tequ-Baijiu* (Grade 1), a high-grade strong-aroma *Baijiu*, employs more of the first fraction during the blending process. *Touqu-Baijiu* (Grade 2), closely trailing in quality, uses more of the middle fraction during its production. *Erqu-Baijiu* (Grade 3), with the lowest quality grade, utilizes more of the final fraction in the blending process (Zheng et al., 2014). During the distillation and blending processes, the classifications depend mainly on the experiences of *Baijiu*-makers and *Baijiu*-panelists, and lack scientific and unified standards. It is difficult to achieve accurate classification of grades, which hinders the development of the *Baijiu* industry.

Molecular sensory science methods are usually based on instrumental analysis (e.g., GC–MS, GC–O, etc.) combined with sensory evaluation, OAV, aroma recombination and deletion experiments, and qualitative and quantitative analyses of the aroma of samples at the molecular level (Guo et al., 2023). At present, molecular sensory science is widely used for the analysis of key aroma compounds in food samples, such as *Baijiu*, *HuangJiu*, fruits, vegetables and meat products (Guo et al., 2023; Wang et al., 2019; Yang et al., 2022; Zhao et al., 2018). Wu et al. (2023a) employed HS-SPME–GC–MS for the analysis of various grades of sauce-aroma *Baijiu*, and found that the quality of *Baijiu* was correlated with the concentrations of aromatics, alcohols, aldehydes, and ketones. Li et al. (2023) used a liquid–liquid extraction (LLE) method combined with GC–O–MS to analyze four different grades of light-aroma *Baijiu*, and found similar results. High-grade *Baijiu* mainly showed pleasant floral, fruity and sweet aromas, while low-grade *Baijiu* mainly showed sour aromas (Li et al., 2023). It was found that compounds such as 3-methylbutyl acetate, 1-butanol, ethyl hexanoate, ethyl butanoate, ethyl octanoate, and acetic acid may be responsible for the grade difference of the four *Baijiu* samples (Li et al., 2023). In addition, pyrazines might be formed by the Maillard reaction at high temperatures during distillation, which impart a nutty aroma to *Baijiu* samples (Zhang et al., 2019).

At present, there is no report on the systematic analysis of different grades of strong-aroma *Baijiu* by comprehensive gas chromatography combined with molecular sensory science. Thus, the purpose of this research is to: (1) analyze the aroma compounds of the three different grades of strong-aroma *Baijiu* by GC×GC–TOF MS, (2) determine the aroma-active compounds by GC–O and AEDA, (3) verify the key aroma-active compounds by recombination and omission experiments, and (4) screen the characteristic aroma compounds of different grades of strong-aroma *Baijiu* by principal component analysis (PCA). The purpose of this study is to provide a theoretical basis for the identification of different grades of strong-aroma *Baijiu* from the perspective of aroma, as well as a scientific basis for the high-quality development of the *Baijiu* industry.

2. Materials and methods

2.1. Materials

The three different grades of strong-aroma *Baijiu* were provided by Luzhou Laojiao Co., Ltd. (Luzhou, Sichuan Province, China). The alcohol volumes of the three samples were 52 %vol (Grade 1), 52 %vol (Grade 2), and 42 %vol (Grade 3), respectively. These samples were stored at 4 °C before analysis.

2.2. Standards and reagents

All authentic standards of the aroma-active compounds used in this study were at least 95 % purity. Chemicals (purity > 99 %), such as dichloromethane, anhydrous sodium sulfate, *n*-alkanes (C_7–C_{30}), 1,2-dichlorobenzene, ethyl (*Z*)-3-hexenoate, tetramethylpyrazine, 2,6,6-trimethyl-2-cyclohexene-1,4-dione, ethyl 2-acetylbutanoate, 5-ethyl-2(5*H*)-furanone, γ-nonalactone, diethyl butanedioate, butanoic acid, propyl butanoate, ethyl butanoate, *p*-cresol, heptanoic acid, hexanal, hexanoic acid, butyl hexanoate, methyl hexanoate, ethyl hexanoate, isobutyl hexanoate, furfural, nonanol, and pentanoic acid were purchased from Sigma-Aldrich Corp (St. Louis, MO). 2,4-Di-*tert*-butyl phenol, ethyl 2-methylbutanoate, vanillin, 6,10-dimethyl-2-undecanone, γ-hexalactone, ethyl 3-phenylpropanoate, phenylacetaldehyde, phenylacetaldehyde diethyl acetal, butyl lactate, ethyl lactate, ethyl pentanoate, propyl octanoate, octyl octanoate, acetic acid, phenethyl acetate, 3-methylbutyl acetate, octyl isobutanoate, and 2-butanol were purchased from J&K Chemical Corp (Beijing, China). Ethyl phenylacetate, phenethyl hexanoate, methyl salicylate, 3-methylbutyl nonanoate, ethyl hexadecanoate, 1-tridecanol, nonyl acetate, and methyl hexadecanoate were purchased from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). Helium (99.9999 % purity) and nitrogen gas (99.9992 % purity) were purchased from Beijing AP BAIF Gases Industry Co., Ltd. (Beijing, China), and liquid nitrogen was obtained from Beijing Xian Heyu Gas Technology Co., Ltd. (Beijing, China). Ultrapure water (18 MΩ·cm in resistivity) was obtained from a Milli-Q purification system (Millipore, Bedford, MA).

2.3. HS-SPME

Extraction of volatile aroma compounds was performed using the HS-SPME method reported by Wang et al. (2020) with slight modifications. In brief, the samples were diluted with ultrapure water to 10 % ethanol by volume. Each diluted *Baijiu* sample (10 mL) was transferred to a 40 mL screw-capped vial, and 2 μL of 1,2-dichlorobenzene (1.31 mg/mL) were added as an internal standard. The vial was placed in a constant-temperature water bath for equilibration at 40 °C for 20 min. The SPME device containing a 50/30 μm CAR/DVB/PDMS (Supelco, Inc., Bellefonte, PA) fiber was inserted into the vial and extracted at 40 °C for 40 min. The fiber with extracts was injected into the GC injector for desorption for 5 min at 250 °C.

2.4. LLE

The volatile aroma compounds of *Baijiu* were extracted by LLE (Song et al., 2020). In brief, 100 mL of diluted *Baijiu* (10 %vol), 50 mL of dichloromethane, and 10 μL of internal standard 1,2-dichlorobenzene (1.31 mg/mL) were mixed in a conical flask, and the mixture was stirred magnetically for 10 min at 1000 rpm. After centrifuging (Hitachi, Japan) for 30 min at 10000 rpm, the extract, containing volatile aroma compounds, was separated by a separating funnel. The *Baijiu* samples were extracted three times in the same way, and the organic phases were combined. Subsequently, the extract was dehydrated over 150 g anhydrous sodium sulfate for 12 h at 4 °C and filtered using filter paper. The volume was then concentrated to 500 μL under a gentle nitrogen stream. The extract of volatile aroma compounds was stored at –40 °C for further

analysis.

2.5. GC×GC–O–TOF MS

The Agilent 8890 GC (Agilent Technologies, Santa Clara, CA) coupled with a time-of-flight mass spectrometer (TOF MS) (Guangzhou Hexin Instrument Co., Ltd., Guangzhou, China) and an olfactory detection port (ODP3; Gerstel GmbH & Co. KG, Mülheim an der Ruhr Germany) was employed to analyze the volatile aroma profile of the three *Baijiu* samples. The GC system consisted of two columns: the primary column was DB-Wax (60 m × 0.25 mm × 0.25 μm, Agilent J&W) and the second column was DB-17 ms (1.85 m × 0.18 mm × 0.18 μm) with a modulator (J&X Technology Co., Ltd, Shanghai, China) placed between the two columns. The helium flow rate was maintained at 2.1 mL/min. The oven temperature was initially set at 40 °C for 5 min, then increased to 230 °C at a rate of 3 °C/min, and maintained for 10 min. The modulator cold zone temperature was set at –50 °C. The temperatures for hot zone entry and exit were set at 70 °C and 160 °C offsets relative to the oven temperatures, which had maximum temperatures of 260 °C and 320 °C respectively. The modulation period was 5 s. For TOF MS detection, the interface and ion source were programmed at 250 °C and 230 °C, respectively, with electron impact ionization at 70 eV, acquisition range (*m/z*) at 35–350 amu, and a scan rate of 100 scans/s. The ODP3 transmission line and exit port were both maintained at 230 °C. Three experienced panelists described the aromas and assessed the strength of each aroma at the olfactory detection port. Only compounds that were perceived by two or three panelists were designated as aroma-active compounds.

2.6. Aroma extract dilution analysis (AEDA)

AEDA was used to characterize the contribution of aroma compounds in *Baijiu*. For LLE, the aroma extract was diluted with dichloromethane in proportions of 1:3, 1:9, 1:27, 1:81, etc. Injection volume was 1 μL in splitless mode. Three experienced panelists were selected for GC-O analysis. If an aroma was detected by two or three panelists, the corresponding aroma-active compound was deemed to be present in the sample. The aroma-active compounds were identified by the FD factor of the last dilution at which the aroma remained perceptible (Sarhir et al., 2019). For SPME, the aroma compounds were diluted by changing the split ratio of the injection port to 1:3, 1:9, 1:27, and 1:81, etc. The other procedures were consistent with the AEDA of LLE (Zheng et al., 2016b).

2.7. Identification of aroma compounds

The peaks were identified by comparing the MS fragmentation patterns with the National Institute of Standards and Technology (NIST) MS Spectral Library (Version 2023) and Wiley Registry 10th MS Spectral Library. Additionally, their aroma descriptions and linear retention indices (LRIs) were compared with the aroma database (The Good Scents Company Information System) and NIST 2023/Wiley Registry 10th, respectively. The aroma compounds identified in the three different grades of samples were shown in Table 1.

2.8. Quantification of aroma compounds

The external standard method was used to quantify the aroma-active compounds in the three *Baijiu* samples, and the quantitative process was similar to the extraction process of the aroma compounds. Briefly, based on the quantitative results of the internal standard, seven mixed standard solutions with different concentration gradients were prepared, which contained compounds such as ethyl acetate, 2-butanol, ethyl butanoate, propyl butanoate, 3-methylbutyl acetate, ethyl pentanoate, ethyl 2-acetylbutanoate, methyl hexanoate, and ethyl hexanoate, etc. The solutions were extracted three times with 50 mL of dichloromethane and concentrated to 500 μL. Selective ion mode (SIM) was used for accurate quantification of the aroma-active compounds.

2.9. Calculation of OAVs

The contribution of aroma compounds could be expressed by the ratio of their concentrations to the thresholds, known as the OAVs. The threshold values of aroma compounds were related to the matrix, which affected the release and perception of aroma compounds. In this study, the OAVs were calculated based on the external standard quantification results and the threshold values of the compounds in ethanol aqueous solutions.

2.10. Aroma recombination and omission experiments

To validate the aroma-active compounds identified by AEDA and OAVs, 27, 29 and 12 aroma compounds (Table 2) were added to an ethanol aqueous solutions (ethanol contents consistent with that of *Baijiu* samples) to create three recombination models, and then, the recombination models were evaluated using 0–5 point scales.

Aroma omission experiments were carried out by means of a triangulation test. Two recombination models without missing compounds and one omission model with one compound missing were randomly numbered and randomly presented, and the panelists selected the different sample out of the three samples (Wang et al., 2023).

2.11. Data processing

All experiments were conducted in triplicate, and the results were presented as the mean ± standard deviation. Tables were created using Microsoft Office Excel 2021 (Microsoft Corp., Redmond, WA), while the bar charts, clustered heat maps, and principal component analysis results were obtained using OriginPro 2023 (OriginLab Corp., Northampton, MA).

3. Results and discussion

3.1. Comparison of the three different grades of strong-aroma Baijiu

Through sensory evaluation by the panelists, a total of 10 aroma attributes were selected, namely fruity, sweet, floral, acidic, grassy, cellar, smoky/burnt, roast/nut, alcoholic, and grain. A quantitative sensory description analysis was conducted on three different grades of *Baijiu*, and the odor profiles of the three samples were obtained (Fig. 1). The sensory scores for fruity and sweet in G1 sample were significantly higher than those in G2 and G3. This may be due to the higher concentrations of ester compounds in G1 sample, which typically impart fruity and sweet aromas to the sample and influence consumer preference (Wang et al., 2022). This is one of the reasons why G1 sample is more favored. Cellar aroma is one of the typical characteristics of strong-aroma *Baijiu* (Dong et al., 2019). As observed, G2 had the strongest cellar aroma, followed by G1, while G3 had the weakest cellar aroma. However, the intense cellar aroma in G2 could overshadow other aroma characteristics, making the overall aroma of the *Baijiu* less harmonious. Grain aroma, also known as cereal aroma, is another important attribute in strong-aroma *Baijiu*. The intensity of grain aroma in G1 and G2 showed no significant difference, while G3 had the weakest grain aroma. G3 sample had the strongest acidic aroma, followed by G2, while G1 had the weakest acidic aroma. However, a stronger acidic aroma can lead to a pungent odor, affecting the quality of *Baijiu*. In general, the profile of G1 sample was more balanced and harmonious. The prominent pit mud aroma in G2 and the pronounced acidic aroma in G3 disrupted the aroma balance, thus affecting their quality.

Table 1
Aroma compounds identified in the three different grades of strong-aroma *Baijiu*.

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
Esters (146)								
1	ethyl acetate	fruity, sweet, green	1100000±43700[a]	27	950000±86000[b]	3	269000±20400[c]	1
2	methyl butanoate	fruity, sweet, pineapple	174±5.60[a]	1	-	-	-	-
3	1-methylpropyl acetate	fruity, banana	855±42.0[b]	1	1120±78.7[a]	1	650±33.5[c]	1
4	ethyl acrylate	fruity	245±15.5[b]	-	330±16.8[a]	1	-	-
5	isobutyl acetate	sweet, fruity, banana	203±21.2[b]	-	328±20.4[a]	1	-	-
6	ethyl butanoate	fruity, pineapple	705000±23700[a]	3	473000±5960[b]	3	203000±289[c]	1
7	ethyl 2-methylbutanoate	sweet, apple	2220±95.5[a]	9	18300±185[b]	1	610±67.9[c]	-
8	ethyl 3-methylbutanoate	fruity, pineapple	3720±187[a]	1	2290±139[b]	1	766±91.9[c]	1
9	butyl acetate	fruity, banana	103±11.3[c]	-	1250±44.5[a]	1	198±11.9[b]	-
10	propyl butanoate	fruity, pineapple	35900±3750[a]	9	756±61.3[b]	-	65.3±6.90[c]	-
11	3-methylbutyl acetate	fruity, banana	-	-	2350±257[a]	81	2.19±0.230[b]	-
12	ethyl pentanoate	apple, pineapple	22300±904[b]	243	25000±1400[a]	729	4980±176[c]	27
13	ethyl 2-methylpentanoate	fruity, pineapple	70.4±9.06[b]	1	76.6±5.85[b]	-	87.5±3.24[a]	-
14	ethyl (*E*)-2-butenoate	caramel, rum	317±18.9[a]	1	298±13.4[a]	1	83.1±6.60[b]	1
15	isobutyl butanoate	pineapple, rum, fruit	41.7±4.23[b]	-	105±9.77[a]	1	-	-
16	pentyl acetate	banana, apple	45.4±8.05[b]	-	197±20.3[a]	1	-	-
17	ethyl 3-methylpentanoate	pineapple, fruity	-	-	-	-	1.17±0.0900[a]	-
18	ethyl 2-acetylbutanoate	fruity, winey, green	39.2±6.75[a]	3	14.4±2.33[c]	1	24.2±2.44[b]	3
19	ethyl 4-methylpentanoate	fruity	1770±127.2[b]	-	2220±75.0[a]	-	179±21.5[c]	-
20	methyl hexanoate	pineapple, strawberry	2480±329[a]	27	1790±167[b]	9	159±13.9[c]	1
21	butyl butanoate	banana, pineapple	591±63.8[a]	-	563±35.4[a]	-	-	-
22	ethyl hexanoate	sweet, pineapple	3270000 ±112000[a]	19683	3000000 ±200000[a]	19683	1480000±14200[b]	2187
23	ethyl pyruvate	fruity, sweet, rum, caramel	1050±28.7[a]	1	1120±115[a]	1	-	-
24	ethyl (*E*)-2-methyl-2-butenoate	fruity, floral, caramel	-	-	-	-	196±15.1[a]	-
25	butyl pentanoate	sweet, pineapple	105±7.10[b]	-	1400±79.8[a]	-	-	-
26	3-methylbutyl butanoate	green, pear	714±25.9[b]	1	945±70.7[a]	1	22.9±2.26[c]	-
27	hexyl acetate	fruity, apple	1280±64.1[b]	1	1710±85.9[a]	1	-	-
28	3-methylbutyl 2-methylbutanoate	citrus, apple	32.1±1.92[b]	-	72.5±4.28[a]	-	2.43±0.300[c]	-
29	allyl 2-ethylbutanoate	fruit	-	-	17.6±0.990[a]	-	3.32±0.230[b]	-
30	ethyl (*E*)-2-hexenoate	fruity, pineapple, cheesy	590±43.4[a]	3	299±26.6[b]	3	261±26.7[b]	1
31	ethyl (*Z*)-3-hexenoate	green, apple	187±22.8[b]	-	547±23.9[a]	9	178±11.0[b]	-
32	ethyl (*E*)-3-hexenoate	sweet, fruity, honey	7.16±0.890[a]	-	2.96±0.450[b]	-	-	-
33	3-methylbutyl 3-methylbutanoate	sweet, green, apple	25.7±2.73[a]	1	-	-	-	-
34	ethyl 2-ethylhexanoate	fruity, herbal	-	-	49.8±3.50[a]	-	4.24±0.480[b]	-
35	propyl hexanoate	pineapple, green	3300±230[b]	1	4660±397[a]	-	502±20.2[c]	-
36	ethyl lactate	fruity, buttery	7580±574[b]	3	14300±503[a]	3	5050±242[c]	3
37	ethyl heptanoate	pineapple, wine	10700±788[b]	1	14200±435[a]	1	4120±213[a]	1
38	ethyl 2-hexenoate	rum, juicy	720±25.5[c]	1	954±57.6[b]	1	1340±97.5[a]	1
39	hexyl isobutanoate	fruity, apple, wine	-	-	145±13.7[a]	-	-	-
40	pentyl 3-methylbutanoate	apple, fresh	-	-	14.6±0.130[a]	-	-	-
41	isobutyl hexanoate	pineapple, green	6360±166[a]	3	5690±179[b]	3	105±4.23[c]	1
42	3-octyl acetate	grapefruit, rose, mint	-	-	-	-	5.31±0.610[a]	-
43	3-methylbutyl pentanoate	apple, green	856±40.8[a]	1	683±29.0[b]	-	34.3±3.02[c]	-
44	allyl hexanoate	pineapple, rum	41.1±6.00[a]	1	-	-	-	-
45	pentyl butanoate	pineapple	-	-	-	-	2.36±0.170[a]	-
46	2-methylbutyl hexanoate	ethereal	993±67.8[a]	1	929±62.4[a]	1	72.5±7.71[b]	1
47	butyl hexanoate	pineapple, apple, wine	5730±224[b]	729	7320±187[a]	729	433±39.7[c]	27
48	hexyl butanoate	fruity, apple	-	-	856±88.5[a]	1	-	-
49	propyl lactate	wine	1950±61.4[a]	1	-	-	-	-
50	propyl heptanoate	pineapple, wine	-	-	5.09±0.450[a]	-	-	-
51	ethyl cyclohexylformate	fruity, wine	124.3±4.41[a]	1	105±4.64[c]	-	13.6±1.84[b]	-
52	isobutyl heptanoate	herbal, fruity	182±20.2[a]	-	189±22.6[a]	-	-	-
53	hexyl 2-methylbutanoate	green, fruity, apple	76.9±5.87[b]	1	95.3±3.14[a]	-	7.10±1.10[c]	-
54	ethyl octanoate	fruity, wine	8510±253[a]	-	8930±278[a]	1	1720±108[c]	-
55	hexyl 3-methylbutanoate	sweet, green, strawberry	272±15.6[a]	-	154±5.21[b]	-	8.38±1.06[c]	-
56	ethyl (*Z*)-4-octenoate	fresh, juicy, pineapple	48.9±8.31[a]	-	28.4±1.91[b]	-	-	-
57	3-methylbutyl hexanoate	fruity, banana, pineapple	2670±26.7[b]	-	3390±256[a]	1	158±16.2[c]	-
58	octyl acetate	green, herbal	247±5.32[a]	-	207±7.29[b]	-	11.9±0.950[c]	-

(*continued on next page*)

Table 1 (*continued*)

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
59	ethyl nonanoate	rose, rum, wine	1750±127^{b}	-	2020±104^{a}	-	68.2±5.70^{c}	-
60	ethyl 2-hydroxy-3-methylbutanoate	pineapple, strawberry	307±28.6^{c}	-	803±24.4^{a}	1	624±83.8^{b}	1
61	(*Z*)-3-hexenyl 2-methylbutanoate	green, fruity	-	-	19.9±2.13^{a}	-	-	-
62	ethyl 3-hydroxybutanoate	fruity, green, grape	76.6±2.61^{b}	-	498±44.4^{a}	-	62.1±7.57^{c}	-
63	pentyl hexanoate	sweet, green, apple	2250±104^{b}	1	3310±236^{a}	1	171±8.97^{c}	-
64	butyl lactate	green, wine	2510±289^{a}	9	2170±240^{a}	1	-	-
65	propyl octanoate	coconut	267±29.9^{b}	-	814±77.4^{a}	9	243±25.9^{b}	-
66	isobutyl octanoate	fruity, green, floral	268±22.0^{a}	-	26.6±3.01^{b}	-	10.3±0.860^{c}	-
67	furfuryl acetate	sweet, fruity, banana	-	-	98.9±9.94^{a}	-	-	-
68	ethyl 2-hydroxy-4-methylpentanoate	fresh, blackberry	4300±419^{b}	1	6280±502^{a}	1	552±12.5^{c}	1
69	methyl 2-hydroxy-4-methylpentanoate	sweet, fruity	-	-	508±50.3^{a}	-	14.5±1.75^{b}	-
70	octyl isobutanoate	green, waxy	-	-	31.5±2.94^{a}	9	-	-
71	ethyl (*E*)-2-octenoate	fruity, green	122±15.5^{a}	1	125±7.71^{a}	-	-	-
72	ethyl (*Z*)-2-octenoate	green, pineapple, juicy	-	-	64.0±9.16^{a}	-	7.55±0.900^{b}	-
73	3-methylbutyl heptanoate	herbal, grassy, fruit	164±13.0^{b}	1	2667±18.4^{a}	1	58.7±6.98^{c}	-
74	3-methylbutyl lactate	fruity, creamy	-	-	2930±179^{a}	-	-	-
75	4-methylpentyl 4-methylpentanoate	fruity, herbal	-	-	153±10.6^{a}	1	-	-
76	hexyl hexanoate	herbal, fresh, vegetable	6950±683^{a}	1	7050±514^{a}	1	914±96.2^{b}	-
77	diethyl malonate	sweet, fruity, green	360±13.7^{b}	-	455±39.9^{a}	-	-	-
78	nonyl acetate	waxy, green, fruity	349±41.9^{b}	1	765±24.6^{a}	9	846±80.2^{a}	1
79	(*Z*)-2-penten-1-yl hexanoate	banana	18.3±2.00^{a}	-	13.3±0.760^{b}	-	-	-
80	γ-pentalactone	herbal, sweet, woody	529±53.7^{b}	-	686±39.2^{a}	-	5.60±0.700^{c}	-
81	ethyl 3-acetylpropanoate	sweet, floral, pineapple	510±17.4^{b}	-	671±23.7^{a}	-	213±11.8^{c}	-
82	methyl 3,7-dimethyl-6-octenoate	waxy, fruity, wine	-	-	1.86±0.290^{a}	-	-	-
83	ethyl 2-furoate	fruity, floral	2410±121^{a}	1	2600±215^{a}	-	11.5±0.900^{b}	-
84	γ-butyrolactone	fatty, caramel	-	-	-	-	5.32±0.510^{a}	-
85	octyl butanoate	waxy, fruity, green	-	-	-	-	100±9.12^{a}	1
86	ethyl decanoate	fruity, apple, brandy	1900±77.5^{b}	1	2650±47.7^{a}	1	120±11.4^{c}	1
87	heptyl pentanoate	fruity	-	-	2.61±0.410^{a}	-	-	-
88	ethyl 3-cyclohexylpropanoate	sweet, pineapple	184±22.3^{a}	-	111±7.65^{b}	-	-	-
89	ethyl benzoate	fruity, musty, sweet	648±62.2^{a}	1	650±63.8^{a}	-	41.7±4.61^{b}	-
90	3-methylbutyl octanoate	sweet, fruity, pineapple	772±51.1^{a}	-	717±55.2^{a}	-	84.3±10.2^{b}	-
91	ethyl (*E*)-4-decenoate	green, fruity, waxy	530±53.4^{b}	-	667±61.7^{a}	-	42.1±3.59^{c}	-
92	diethyl butanedioate	fruity, apple	2370±125^{b}	9	4340±162^{a}	1	388±18.9^{c}	3
93	ethyl 3-hydroxyhexanoate	grape, pineapple	31200±2220^{a}	-	23300±78.0^{b}	1	2210±144^{c}	1
94	(*Z*)-3-hexenyl hexanoate	green, wine, grassy	29.5±5.53^{a}	-	-	-	-	-
95	heptyl hexanoate	green, fresh	803±90.2^{a}	1	245±16.6^{b}	1	27.0±3.95^{c}	-
96	γ-hexalactone	herbal, sweet	1350±41.6^{a}	1	303±15.7^{c}	1	551±30.2^{b}	27
97	ethyl 4-acetylbutanoate	fresh	-	-	-	-	132±12.8	-
98	heptyl heptanoate	grassy, green	-	-	682±34.4^{a}	-	-	-
99	propyl decanoate	fatty, green, fruity	52.2±7.55^{a}	-	63.2±5.52^{a}	-	16.2±1.76^{b}	-
100	geranyl acetate	floral, rose, green	-	-	1.60±0.240^{a}	-	-	-
101	ethyl undecanoate	waxy, fatty, coconut	60.8±9.89^{b}	-	89.6±9.52^{a}	-	50.9±3.97^{c}	-
102	isobutyl decanoate	sweet, brandy	-	-	30.7±2.25^{a}	-	-	-
103	ethyl (*E*)-2-decenoate	green, fatty, herbal	180±24.8^{a}	-	-	-	-	-
104	3-methylbutyl nonanoate	fruity, floral	16.8±2.68^{a}	1	17.0±1.25^{a}	3	1.62±0.270^{b}	-
105	methyl salicylate	mint	-	-	138±9.01^{a}	9	-	-
106	ethyl phenylacetate	sweet, floral, honey, rose	1660±174^{b}	9	2150±87.3^{a}	1	220±17.9^{c}	-
107	isobutyl benzoate	sweet, fruity	3.21±0.660^{b}	-	6.09±1.01^{a}	-	-	-
108	δ-hexalactone	creamy, fruity, coconut	60. 8±5.69^{a}	-	-	-	46.2±6.09^{b}	-
109	ethyl 10-undecenoate	fruity, creamy	33.8±5.29^{b}	-	72.0±7.54^{a}	-	-	-
110	γ-heptalactone	sweet, nutty, caramel	494±31.2^{b}	1	738±52.4^{a}	1	506±22.5^{b}	1
111	phenethyl acetate	rose, sweet, honey	231±16.5^{b}	243	308±12.1^{a}	729	11.4±0.280^{c}	81
112	hexyl octanoate	green, apple	1440±152^{a}	1	1620±81.9^{a}	1	493±20.1^{b}	-
113	butyl decanoate	whiskey	26.5±4.19^{b}	1	38.3±5.63^{a}	-	-	-
114	ethyl laurate	sweet, waxy, floral	626±5.88^{b}	-	783±40.3^{a}	-	59.5±5.11^{c}	-
115	furfuryl hexanoate	green, fatty, waxy	255±30.9^{a}	-	243±14.1^{a}	-	43.6±5.49^{b}	-
116	3-methylbutyl decanoate	banana, sweet, green	-	-	27.4±2.75^{a}	-	-	-
117	ethyl 3-phenylpropanoate	rose, honey, rum	1360±20.9^{b}	1	1580±22.3^{a}	3	16.3±1.74^{c}	1
118	trans-3-methyl-4-octanolide	coconut, clove	26.04±3.46^{a}	-	-	-	-	-
119	3-methylbutyl benzoate	sweet, green	30.5±10.0^{a}	1	32.6±4.07^{a}	-	-	-

(*continued on next page*)

Table 1 (*continued*)

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
120	γ-octalactone	coconut, creamy, dairy	72.0±4.58^{b}	-	136±14.1^{a}	-	-	-
121	menthyl 3-methylbutanoate	fruity, sweet	-	-	372±34.9^{a}	1	-	-
122	phenethyl butanoate	floral, strawberry	75.2±14.5^{b}	1	115±5.47^{a}	1	-	-
123	ethyl 4-phenylbutanoate	sweet, fruity	14.8±2.01^{b}	-	23.1±3.33^{a}	-	-	-
124	octyl octanoate	coconut, fruity	-	-	6090±205^{a}	3	-	-
125	isopropyl tetradecanoate	oily, fatty	1.77±0.440^{b}	-	5.67±1.08^{a}	-	1.82±0.320^{b}	-
126	ethyl tetradecanoate	sweet, waxy	549±52.0^{a}	1	656±43.8^{a}	1	148±5.02^{b}	-
127	phenethyl 3-methylbutanoate	floral, fruity, sweet	16.1±3.09^{a}	1	-	-	-	-
128	phenethyl pentanoate	fruity, rose, leaf	-	-	27.5±3.09^{a}	-	4.40±0.440^{b}	-
129	benzyl hexanoate	green, fruity	8.90±2.15^{a}	-	-	-	-	-
130	ethyl (*E*)-cinnamate	floral, honey	3.67±0.980^{a}	-	-	-	1.94±0.180^{b}	-
131	ethyl (*Z*)-cinnamate	sweet, fruity	-	-	4.70±0.310^{a}	9	-	-
132	diisobutyl hexanedioate	mild, estery	35.6±3.95^{b}	1	91.3±11.0^{a}	-	8.48±1.40^{c}	-
133	hexyl phenylacetate	green, rose, wine	3.09±0.450^{b}	-	5.67±0.870^{a}	-	-	-
134	ethyl pentadecanoate	honey, sweet	38.8±2.82^{a}	-	45.2±6.07^{a}	-	5.91±1.12^{b}	-
135	phenethyl hexanoate	honey, floral, sweaty	111±13.6^{b}	3	658±21.5^{a}	1	16.3±2.00^{c}	3
136	diethyl nonanedioate	sweet	230±21.4^{a}	-	5.36±0.830^{b}	-	-	-
137	methyl hexadecanoate	oily, waxy, fatty	54.7±2.20^{c}	1	89.7±10.1^{a}	1	54.3±4.91^{b}	27
138	γ-undecanolactone	peach, creamy, fatty	-	-	257±10.7^{a}	-	-	-
139	ethyl hexadecanoate	waxy, creamy, milky	4120±147.7^{b}	9	7370±292^{a}	1	463±46.4^{c}	-
140	ethyl 2-hydroxy-3-phenylpropanoate	black pepper	1190±200^{a}	1	794±60.1^{b}	1	210±19.1^{c}	-
141	diethyl *o*-benzenedicarboxylate	odorless	-	-	27.6±3.69^{a}	-	-	-
142	methyl octadecanoate	oily waxy	-	-	-	-	17.6±0.850^{a}	-
143	ethyl ocatadecanoate	waxy	54.5±3.98^{a}	-	59.3±5.78^{a}	-	13.7±1.82^{b}	-
144	ethyl (*Z*)-9-octadecenoate	fatty, milk	2090±71.6^{b}	-	2920±115^{a}	-	182±11.6^{c}	-
145	methyl (*Z*)-9-octadecenoate	fatty	-	-	521±28.4^{a}	-	-	-
146	ethyl linoleate	fatty, fruity	588±37.3^{b}	-	1900±129^{a}	1	-	-
Content of total esters			5280000 ±120000^{a}		4630000 ±287000^{b}		1980000±34600^{c}	
Carboxylic acids (17)								
1	acetic acid	sour, vinegar	1620±1580^{b}	1	24300±2080^{a}	9	10300±651^{c}	1
2	propionic acid	pungent, acidic	2950±47.7^{a}	1	1950±178^{b}	-	1040±73.9^{c}	-
3	isobutanoic acid	acidic, sour, cheese	35100±428^{a}	1	26400±703^{b}	1	3650±298^{a}	-
4	(*E*)-2-nonenoic acid	fatty, oily	21.4±3.62^{a}	-	-	-	-	-
5	butanoic acid	acetic, cheese, butter	86400±9130^{a}	81	597000±4510^{b}	729	273000±1070^{c}	27
6	pentanoic acid	acidic, sweaty, rancid	68100±4300^{a}	3	52300±1600^{b}	1	12500±190^{c}	1
7	3-methylbutanoic acid	feet, sweaty, cheese	9850±725^{a}	1	11700±986^{a}	1	2700±91.9^{b}	1
8	pyruvic acid	sour	-	-	1.23±0.240^{a}	-	-	-
9	2-methylpentanoic acid	sour, cheese	716±62.2^{a}	-	-	-	-	-
10	3-methylpentanoic acid	acidic, cheesy, sweaty	-	-	370±14.9^{a}	-	-	-
11	4-methylpentanoic acid	pungent, cheese	235±17.8^{a}	-	-	-	101±6.51^{b}	-
12	2-methylhexanoic acid	cheese, fatty	358±42.9^{a}	1	-	-	41.0±3.64^{b}	1
13	hexanoic acid	sour, fatty, sweat	1230000±33000^{a}	729	914000±13900^{b}	729	851000±26000^{c}	27
14	heptanoic acid	rancid, sour, sweat	22300±389^{a}	3	9350±879^{b}	1	6470±112^{c}	3
15	octanoic acid	fatty, rancid	15300±440^{a}	1	9380±270^{b}	1	1210±67.7^{c}	-
16	nonanoic acid	waxy, cheese	52.2±1.93^{a}	1	60.6±9.50^{a}	1	-	-
17	decanoic acid	rancid, sour, fatty	-	-	95.7±9.12^{a}	-	-	-
Content of total carboxylic acids			2260000±40700^{a}		1650000±21000^{b}		1160000±26600^{c}	
Alcohols (53)								
1	2-butanol	sweet, apricot	3340±114^{b}	27	4310±256^{a}	27	1290±105^{c}	-
2	1-propanol	alcoholic, musty	3420±144^{b}	-	4530±384^{a}	1	1100±65.5^{c}	-
3	isobutanol	winey	3990±158^{b}	1	4810±458^{a}	-	341±20.9^{c}	-
4	2-propen-1-ol	pungent	-	-	1.06±0.200^{a}	-	-	-
5	2-pentanol	green, fermented	2450±278^{b}	1	3360±325^{a}	1	168±3.24^{c}	-
6	1-butanol	sweet, whiskey	8300±298^{b}	1	10100±751^{a}	1	1350±54.1^{c}	-
7	3-methyl-2-butanol	fruity	-	-	-	-	13.6±2.21^{a}	-
8	3-methyl-3-pentanol	fruity, green, leafy	-	-	50.0±2.11^{b}	-	73.2±5.27^{a}	-
9	3-methyl-1-butanol	alcoholic, whiskey, fruity	27900±3060^{a}	1	21500±1910^{b}	-	1010±53.3^{c}	-
10	2-hexanol	winey, fruity, fatty	591±9.16^{a}	-	649±62.0^{a}	-	23.9±3.06^{c}	-
11	1-pentanol	fusel, oil, sweet	30200±1490^{a}	1	23700±1100^{b}	-	897±66.1^{c}	-
12	cyclopentanol	sesame	260±30.2^{b}	1	353±22.1^{a}	-	-	-
13	(*Z*)-2-penten-1-ol	green, cherry, metallic	189±5.94^{a}	-	67.8±8.24^{b}	-	-	-
14	2-heptanol	grass, herbal, sweet	555±23.8^{b}	-	636±43.6^{a}	-	52.5±5.39^{c}	-
15	1-hexanol	fruity, sweet, green	57900±470^{a}	1	39900±1230^{b}	1	2230±23.7^{c}	-
16	6-methyl-2-heptanol	waxy, fatty, citrus	-	-	74.2±6.37^{a}	-	-	-

(*continued on next page*)

Table 1 (*continued*)

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
17	(*Z*)-3-hexen-1-ol	green, grass, herbal	-	-	-	-	33.5±3.17^{a}	1
18	3-ethoxy-1-propanol	fruit	334±32.6^{b}	1	462±18.3^{a}	-	-	-
19	(*E*)-3-hexen-1-ol	green, leafy, floral	183±6.27^{a}	-	-	-	55.0±4.43^{b}	-
20	3-octanol	mushroom, herbal	-	-	105±8.78^{a}	-	18.8±2.74^{b}	-
21	2-octanol	woody, herbal	90.7±10.2^{b}	-	603±43.21^{a}	-	61.6±6.04^{c}	-
22	1-octen-3-ol	mushroom	265±17.6^{a}	1	227±18.5^{a}	1	-	-
23	6-methyl-5-hepten-2-ol	sweet, green	-	-	1.50±0.220^{a}	-	-	-
24	2-ethyl-1-hexanol	citrus, fresh	380±49.7^{a}	1	390±40.6^{a}	-	229±1.95^{b}	-
25	(*E*)-2-hepten-1-ol	fatty, green	-	-	102±6.20^{a}	-	-	-
26	2-nonanol	green, creamy, orange	-	-	145±16.2^{a}	-	-	-
27	2,6-dimethyl-4-heptanol	ethereal, fermented	-	-	113±10.5^{a}	-	-	-
28	linalool	citrus, woody	10.1±1.60^{a}	-	11.9±1.08^{a}	-	-	-
29	1-octanol	green, orange	1720±98.0^{a}	-	398±21.8^{b}	-	116±8.53^{b}	-
30	2,3-butanediol	fruity, creamy	868±40.7^{b}	1	1250±89.4^{a}	1	430±33.8^{c}	1
31	1,2-propanediol	alcoholic	450±4.59^{a}	-	-	-	-	-
32	(*E*)-2-octen-1-ol	green, citrus	-	-	15.4±2.25^{a}	-	-	-
33	(*Z*)-5-octen-1-ol	green, leaf	-	-	9.68±0.830^{a}	1	-	-
34	2-furanmethanol	caramel, bread	1700±71.6^{b}	-	2240±303^{a}	1	670±70.5^{c}	1
35	1-nonanol	fatty, floral, rose	210±24.0^{a}	9	226±19.6^{a}	81	12.5±0.710^{b}	-
36	(*Z*)-3-nonen-1-ol	green, mushroom	10.5±1.10^{b}	1	14.0±1.76^{a}	-	-	-
37	(*E*)-2-nonen-1-ol	waxy, green	-	-	18.0±0.340^{a}	-	-	-
38	(*Z*)-2-nonen-1-ol	fatty, cucumber	14.6±1.74^{b}	-	18.6±1.22^{a}	-	1.69±0.190^{c}	-
39	2-undecanol	fresh, waxy	124±7.57^{a}	-	120±8.61^{a}	-	10.7±0.980^{b}	-
40	2-phenyl-2-propanol	green, earthy	-	-	-	-	2.52±0.350^{a}	-
41	1-decanol	fatty, waxy, orange	79.8±5.54^{a}	-	77.1±9.44^{a}	-	36.2±3.99^{b}	1
42	(*Z*)-4-decen-1-ol	waxy, fatty	-	-	18.0±1.55^{a}	-	-	-
43	(*Z*)-2-decen-1-ol	citrus, rose	-	-	4.47±0.410^{a}	-	-	-
44	(*E*)-5-decen-1-ol	waxy, floral	-	-	-	-	38.0±1.74^{a}	-
45	(*E*)-2-decen-1-ol	waxy, citrus	-	-	-	-	4.98±0.260^{a}	-
46	phenylmethanol	floral, rose	-	-	477±34.5^{a}	-	-	-
47	phenylethanol	rose, flower	4330±201^{a}	1	4010±134^{a}	1	707±20.0^{b}	1
48	(*E*)-2-undecen-1-ol	floral, fruity	9.94±1.12^{a}	-	-	-	-	-
49	1-dodecanol	waxy, fatty	-	-	2380±251^{a}	1	1570±3.72^{b}	1
50	(*Z*)-nerolidol	floral	-	-	-	-	1.99±0.260^{a}	-
51	(*E*)-nerolidol	floral, citrus	14.6±1.40^{b}	-	28.2±1.35^{a}	-	-	-
52	phenylpropanol	sweet	50.6±7.33^{a}	-	-	-	-	-
53	1-tridecanol	musty	-	-	6.81±0.970^{a}	27	-	-
Content of total alcohols			150000±3380^{a}		128000±4230^{b}		12500±238^{c}	
Aldehydes (31)								
1	2-butenal	flower	4.76±0.430^{b}	-	78.2±8.83^{a}	1	-	-
2	hexanal	grass, green, fatty	1460±204^{b}	1	1900±166^{a}	3	167±17.4^{c}	-
3	(*E*)-2-methyl-2-butenal	green, fruit	-	-	-	-	21.0±2.55^{a}	-
4	3-methyl-2-butenal	sweet, fruity, nutty	-	-	-	-	17.4±2.27^{a}	-
5	nonanal	rose, fresh	693±11.3^{b}	-	60.0±1.96^{c}	-	861±3.48^{a}	-
6	(*E*)-2-octenal	cucumber, leaf	52.0±4.08^{b}	-	129±9.30^{a}	-	5.13±0.750^{c}	-
7	furfural	sweet, bread	72000±3280^{a}	81	53800±1530^{b}	9	12400±107^{c}	1
8	(*E,E*)−2,4-heptadienal	fatty, green	-	-	4.66±0.920^{a}	-	-	-
9	decanal	waxy, citrus	135±18.3^{a}	1	118±1.96^{ab}	1	106±10.4^{b}	1
10	benzaldehyde	almond	826±31.7^{a}	1	778±68.7^{a}	-	441±6.00^{b}	-
11	(*Z*)-2-nonenal	fatty, cucumber	167±12.7^{a}	1	46.5±4.14^{b}	-	14.4±1.40^{c}	1
12	(*E*)-2-nonenal	green, cucumber	130±9.12^{b}	1	170±13.2^{a}	1	9.77±1.00^{c}	-
13	5-methylfurfural	caramel	98.0±13.1^{c}	-	126±19.7^{b}	-	270±20.8^{a}	-
14	(*E,Z*)-2,6-nonadienal	green, fatty, cucumber	-	-	1.64±0.220^{a}	-	-	-
15	undecanal	citrus, green, fatty	40.6±3.63^{a}	1	46.8±4.46^{a}	-	1.54±0.180^{b}	-
16	3-methylbenzaldehyde	sweet, fruity	-	-	1.75±0.220^{a}	-	-	-
17	*β*-cyclocitral	rose, sweet, fruity	14.6±2.26^{a}	-	15.9±1.73^{a}	-	-	-
18	phenylacetaldehyde	floral, honey	3250±165^{b}	3	3760±229^{a}	81	71.2±4.84^{c}	1
19	(*Z*)-2-decenal	tallow	55.9±6.27^{a}	-	65.4±7.44^{a}	-	-	-
20	(*E*)-2-decenal	waxy, fatty	-	-	33.2±4.70^{a}	-	4.36±0.450^{b}	-
21	2-formylthiophene	sulfurous	454±60.0^{b}	-	747±46.0^{a}	-	3.64±0.490^{c}	-
22	2,4-nonadienal	fatty, green, cucumber	-	-	14.7±0.680^{a}	-	-	-
23	4-ethylbenzaldehyde	bitter, almond	10.9±0.760^{b}	-	19.2±2.84^{a}	-	3.95±0.440^{c}	-
24	dodecanal	waxy, citrus, green	15.4±2.09^{b}	1	28.3±2.83^{a}	-	2.41±0.0900^{c}	-
25	(*E*)-2-undecenal	fresh, fruity, citrus	-	-	-	-	3.75±0.840^{a}	-
26	(*Z*)-2-undecenal	fresh, orange	21.6±3.15^{b}	-	47.0±6.08^{a}	-	-	-
27	tridecanal	soapy, citrus	16. 5±3.71^{a}	1	14.2±1.89^{a}	1	1.29±0.050^{b}	-
28	tetradecanal	waxy, citrus	-	-	48.6±3.46^{a}	1	-	-
29	2-phenyl-2-buten-1-al	beany, nutty	53.5±0.320^{b}	-	81.5±7.14^{a}	-	-	-
30	lilac aldehyde	flowery, fresh	-	-	84.5±4.88^{a}	-	-	-
31	vanillin	sweet, vanilla	272±36.6^{a}	6561	-	-	-	-

(*continued on next page*)

Table 1 (*continued*)

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
Content of total aldehydes			79800±3220^{a}		62200±1400^{b}		14400±121^{c}	
Ketone (33)								
1	3-penten-2-one	fruity, fishy	-	-	573±39.1^{a}	-	14.3±1.70^{b}	-
2	cyclopentanone	minty	150±5.16^{a}	-	-	-	-	-
3	2-heptanone	fruity, coconut	7490±48.7^{a}	-	-	-	-	-
4	3-hydroxy-2-butanone	sweet, buttery, creamy	2750±122^{b}	-	3650±126^{a}	1	754±56.8^{c}	1
5	1-hydroxy-2-propanone	sweet, caramellic	262±11.58^{a}	1	199±8.04^{b}	1	42.0±2.70^{c}	1
6	2-octanone	earthy, woody, herbal	236±9.32^{b}	-	146±17.9^{c}	-	334±11.58^{a}	1
7	6-methylhept-5-en-2-one	citrus, green, grass	-	-	9.46±0.810^{b}	-	137±12.9^{a}	-
8	2-hydroxy-3-pentanone	earthy, nutty	721±7.10^{a}	-	-	-	-	-
9	3-nonanone	fresh, sweet, leaf	6.85±1.20^{b}	-	13.5±1.68^{a}	-	-	-
10	3-acetoxy-2-butanone	sweet, creamy, buttery	-	-	49.4±2.97^{a}	-	-	-
11	2-nonanone	green, earthy, herbal	536±33.0^{a}	1	446±14.7^{b}	-	386±20.4^{c}	-
12	3-octen-2-one	herbal, mushroom	12.4±1.54^{a}	-	16.3±2.73^{a}	-	1.89±0.290^{b}	-
13	3-decanone	waxy, fruity	-	-	-	-	21.3±2.04^{a}	-
14	2-decanone	orange, floral	310±22.1^{a}	1	221±18.9^{b}	1	301±30.2^{a}	1
15	3-nonen-2-one	berry, fatty	-	-	3.39±0.190^{a}	-	2.47±0.170^{b}	-
16	4-undecanone	fruity	328±35.6^{a}	-	-	-	-	-
17	6-methyl-3,5-heptadien-2-one	coconut, woody, sweet	-	-	93.7±9.55^{a}	-	-	-
18	isophorone	woody, green, fruity	449±8.07^{a}	-	135±13.2^{b}	-	-	-
19	2-undecanone	waxy, fruity, creamy	158±21.1^{a}	1	188±18.6^{a}	1	36.4±2.93^{b}	1
20	acetophenone	sweet, pungent, almond	-	-	-	-	10.9±1.65^{a}	-
21	2,6,6-trimethyl-2-cyclohexene-1,4-dione	musty, woody, leaf	65.7±8.00^{b}	9	133±12.0^{a}	-	-	-
22	6,10-dimethyl-2-undecanone	musty	-	-	242±25.1^{a}	3	18.6±1.16^{b}	-
23	2-octen-4-one	mushroom, metal	86.9±8.00^{a}	1	-	-	-	-
24	2-dodecanone	citrus	11.9±1.85^{b}	-	23.5±3.31^{a}	-	3.01±0.150^{c}	-
25	1-phenyl-1-butanone	walnut, hazelnut	-	-	17.3±2.17^{a}	-	3.99±0.550^{b}	-
26	2-tridecanone	fatty, waxy	88.4±8.77^{b}	-	128±9.92^{a}	-	17.9±2.11^{c}	-
27	(*E*)-*β*-damascenone	apple, rose, honey	39.6±6.30^{b}	-	52.1±6.42^{a}	-	4.46±0.310^{c}	1
28	(*E*)-6,10-dimethyl-5,9-undecadien-2-one	green, fruity, waxy, rose	142±21.5^{b}	-	213±16.4^{a}	-	-	-
29	(*E*)-*β*-ionone	floral, orris	-	-	8.27±0.780^{a}	-	-	-
30	2-pentadecanone	fresh, jasmin	34.8±6.28^{b}	1	45.0±1.69^{a}	-	9.57±1.76^{c}	-
31	dihydro-5-pentyl-2(3*H*)-furanone	coconut, creamy, buttery	210±4.17^{a}	9	202±5.30^{b}	3	32.5±0.860^{c}	3
32	cyclopentadecanone	fatty	33.3±5.13^{b}	-	92.8±12.3^{a}	-	-	-
33	6,10,14-trimethyl-2-pentadecanone	herbal, woody	8.70±0.680^{b}	-	91.4±10.2^{a}	1	7.34±0.810^{b}	-
Content of total ketones			7390±179^{a}		6990±174^{b}		2140±86.3^{c}	
Terpenes (15)								
1	*α*-copaene	woody, honey	18.1±3.52^{a}	1	-	-	-	-
2	*α*-gurjunene	wood, balsam	-	-	107±4.54^{a}	-	-	-
3	isolongifolene	woody	323±16.8^{a}	-	-	-	-	-
4	*α*-santalene	woody	15.5±1.85^{a}	-	2.91±0.630^{b}	-	-	-
5	longifolene	sweet, rose	1530±200^{a}	-	156±16.9^{b}	-	13.4±1.20^{c}	-
6	(*E*)-*α*-bergamotene	woody	-	-	4.75±0.520^{a}	-	-	-
7	*β*-caryophyllene	woody, clove	134±12.4^{a}	-	-	-	-	-
8	*β*-elemene	sweet	-	-	-	-	0.740±0.0600^{a}	-
9	*β*-santalene	woody	-	-	2.29±0.230^{a}	-	-	-
10	*α*-humulene	woody	17.4±3.29^{a}	-	2.83±0.390^{b}	-	-	-
11	*γ*-muurolene	herbal, woody	29.5±1.57^{a}	-	-	-	-	-
12	aromadendrene	wood	6.41±0.620^{a}	-	-	-	-	-
13	*δ*-cadinene	herbal, woody	41.5±5.20^{a}	-	-	-	3.35±0.440^{b}	-
14	*α*-curcumene	herbal	35.4±4.50^{a}	-	11.6±1.98^{b}	-	-	-
15	calamenene	herbal	102±8.05^{a}	-	-	-	-	-
Content of total terpenes			2250±210^{a}		288±17.8^{b}		17.5±1.58^{c}	
Phenols (8)								
1	2,6-di-tert-butyl-4-methylphenol	phenolic, camphor	208±17.9^{b}	-	252±13.3^{a}	-	155±9.22^{c}	-
2	4-methyl-2-methoxyphenol	vanilla, medicinal, smoky	-	-	-	-	49.3±3.96^{a}	-
3	phenol	phenolic, plastic, rubber	1500±81.9^{b}	-	2400±18.8^{a}	-	1270±126^{c}	-
4	*p*-cresol	phenolic	2270±223^{a}	9	137±19.8^{c}	-	1600±111^{b}	1
5	*m*-cresol	woody, phenolic	-	-	3350±101^{a}	-	-	-
6	2-methoxy-4-vinylphenol	clove, phenolic, peppery	-	-	56.0±7.38^{a}	-	19.0±2.45^{b}	-
7	4-*tert*-butyl phenol	leather	-	-	-	-	44.3±5.42^{a}	-
8	2,4-di-*tert*-butyl phenol	phenolic	3730±207^{a}	3	3150±240^{b}	1	3190±265^{b}	3

(*continued on next page*)

Table 1 (*continued*)

No.	Compound	Aroma description	Grade 1		Grade 2		Grade 3	
			Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor	Concentration/ (μg/L)	FD Factor
Content of total phenols			7710±230^{b}		9340±336^{a}		6330±283^{c}	
Acetals (5)								
1	3-methylbutanal diethyl acetal	fruity, fatty	7330±133^{a}	1	7420±559^{a}	1	892±50.4^{b}	-
2	hexanal diethyl acetal	floral, fruity	294±6.05^{a}	1	-	-	53.6±4.54^{b}	-
3	3-ethoxypropionaldehyde diethyl acetal	vegetable	306±28.8^{a}	-	266±24.9^{a}	1	100±8.52^{b}	1
4	nonanal diethyl acetal	floral, rose	29.9±3.62^{a}	-	-	-	13.1±2.05^{b}	-
5	phenylacetaldehyde diethyl acetal	fresh, green, sweet	2120±27.4^{a}	1	792±90.2^{b}	1	262±26.2^{c}	3
Content of total acetals			10100±195^{a}		8480±479^{b}		1320±70.7^{c}	
Heterocycles (13)								
1	2-vinylfuran	phenolic, coffee	-	-	2.99±0.280^{a}	-	-	-
2	2,6-dimethylpyrazine	roasted, coffee	-	-	-	-	57.1±7.35^{a}	-
3	2-ethyl-6-methylpyrazine	roasted, coffee	-	-	6.56±1.02^{a}	-	-	-
4	trimethylpyrazine	nutty, cocoa, peanut	310±29.0^{a}	-	248±23.2^{b}	-	39.6±6.11^{c}	-
5	2-methyl-2-vinyl-5-(*α*-hydroxy-isopropyl)-tetrahydrofuran	floral, sweet	127±2.41^{a}	-	-	-	-	-
6	2-(diethoxymethyl)-furan	fruity, mushroom	1240±57.9^{a}	-	188±4.23^{c}	-	783±96.5^{b}	-
7	tetramethylpyrazine	nutty, chocolate	603±37.8^{a}	-	648±40.6^{a}	-	75.7±4.80^{b}	3
8	2-acetylfuran	cocoa, caramel	642±51.7^{a}	-	40.7±4.09^{b}	-	-	-
9	2,3-benzofuran	plastic	-	-	3.27±0.320^{a}	-	-	-
10	2-propanoylfuran	fruity	16.9±1.83^{b}	-	16.0±1.10^{b}	-	43.2±5.37^{a}	-
11	2-acetyl-5-methylfuran	nutty, coconut	231±18.5^{a}	-	7.67±0.850^{c}	-	47.4±6.45^{b}	-
12	2(5*H*)-furanone	buttery	-	-	-	-	287±17.7^{a}	-
13	5-ethyl-2(5*H*)-furanone	spice	-	-	-	-	64.5±6.87^{a}	3
Content of total heterocycles			3170±39.8^{a}		1160±58.4^{c}		1400±125^{b}	
Other compounds (21)								
1	dimethyl disulfide	sulfurous, onion	14.2±2.56^{b}	-	45.9±3.75^{a}	-	5.04±0.890^{c}	-
2	styrene	plastic	1220±79.0^{a}	-	603±51.8^{b}	-	82.5±9.68^{c}	-
3	*p*-isopropyltoluene	woody, spice	94.6±12.1^{a}	-	-	-	-	-
4	dimethyl trisulfide	sulfurous, onion	87.7±3.27^{c}	-	192±12.3^{a}	1	106±0.600^{b}	-
5	1,2,4,5-tetramethylbenzene	rancid	107±14.9^{b}	-	348±47.1^{a}	-	27.2±1.95^{c}	-
6	1-methoxy-4-methylbenzene	rancid	-	-	12.1±0.990^{a}	-	-	-
7	1-methyl-4-isopropenylbenzene	phenolic, spicy	-	-	11.9±1.44^{a}	-	-	-
8	butyric anhydride	butter	-	-	12.1±1.83^{a}	-	10.2±0.260^{a}	-
9	camphor	camphor	-	-	-	-	3.30±0.470^{a}	-
10	vitispirane	floral, fruity, woody	81.2±5.90^{a}	-	-	-	-	-
11	2-methoxy-1-methyl-4-(1-methylethyl)-benzene	herbal, leafy, citrus	-	-	1.95±0.270^{a}	-	-	-
12	benzonitrile	rancid	30.0±2.79^{b}	-	112±8.02^{a}	1	3.85±0.540^{c}	-
13	carbitol	ether	18.7±2.14^{a}	-	13.8±1.51^{b}	-	11.4±0.750^{b}	-
14	decanenitrile	fatty, waxy	7.43±0.510^{c}	-	19.8±3.02^{b}	-	39.2±4.16^{a}	-
15	naphthalene	pungent	23.1±3.51^{b}	-	26.4±0.490^{b}	-	59.4±6.56^{a}	-
16	1,1,6-trimethyl-2*H*-naphthalene	licorice	38.8±2.77^{b}	-	54.9±6.60^{a}	-	3.32±0.550^{c}	-
17	safrole	woody	96.0±16.8^{a}	1	71.9±5.82^{b}	-	-	-
18	biphenyl	pungent, geranium	2.15±0.380^{b}	-	5.66±0.350^{a}	-	1.85±0.270^{b}	-
19	isosafrole	spicy	-	-	-	-	1.20±0.290^{a}	-
20	longiborneol	woody	-	-	4.24±0.390^{a}	-	-	-
21	4-vinylphenol	phenolic	-	-	64.2±9.38^{a}	-	-	-
Content of total other compounds			1820±65.6^{a}		1600±39.9^{b}		355±9.44^{c}	
Total identified/detected			7800000 ±151000^{a}		6490000 ±261000^{b}		3180000±58200^{c}	

The different letters marked with a, b and c on the compound concentrations represent significant differences between the data, $p < 0.05$, and there is no significant difference between the data marked with the same letters.

3.2. Aroma compounds in three different grades of Baijiu

As shown in Table 1 and Fig. 2, a total of 342 aroma compounds were identified in three different grades of strong-aroma *Baijiu*. Of these, 146 were esters, which were the most abundant in the three samples, followed by 17 acids, 53 alcohols, 31 aldehydes, 33 ketones, 15 terpenes, 8 phenols, 5 acetals, 13 heterocycles, and 21 other kinds of compounds.

Esters are important aroma fractions in *Baijiu*, which have important influences on the flavor and quality of *Baijiu*. For all three grades of *Baijiu* samples, the quantity and concentration of ester compounds were the highest. A total of 110, 127 and 89 ester compounds were identified in the three samples, and their mean concentrations were 5280, 4630 and 1980 mg/L, respectively. The ester contents in G1 and G2 were similar, but they were more than twice the ester content in G3 sample. Esters usually have pleasant floral, sweet, and fruity aromas (Xu et al., 2022). In addition, they can relax the human body and reduce the side effects caused by drinking (Hossain et al., 2002). The content of ethyl hexanoate was the highest among the ester compounds in the three samples, at 3270, 3000, and 1480 mg/L, respectively. It accounted for 60%–75% of all esters, showing a sweet, pineapple and green aroma (Wei et al., 2020). Ethyl hexanoate was mainly produced by the esterification of hexanoic acid (generated by hexanoic acid-producing bacteria) and ethanol under the action of esterase in the later stage of fermentation (Gawas et al., 2018; Xu et al., 2021). Ethyl hexanoate has a cellar aroma, which is one of the characteristics of high-quality strong-aroma *Baijiu*. Ethyl acetate was the second most abundant compound in all three samples, with concentrations of 1100, 950, and 269 mg/L in G1, G2, and G3, respectively, imparting a fruity and sweet aroma to the samples. The ethyl acetate content in G1 and G2 was approximately 3.5–4 times higher than that in G3. The content of ethyl butanoate ranked third among the esters in the three samples, with a pineapple and banana aroma (Xu et al., 2020). It had the highest content

Table 2
Concentrations and odor-activity values of odor-active compounds in the three different grades of *Baijiu*.

No.	Compound	Quantitive ions (*m/z*)	Linear equation	R^2	Threshold/ (μg/L)	C_{G1}/(μg/L)[a]	FD_{G1}[b]	OAV_{G1}[c]	C_{G2}/(μg/L)[d]	FD_{G2}[e]	OAV_{G2}[f]	C_{G3}/(μg/L)[g]	FD_{G3}[h]	OAV_{G3}[i]
1	ethyl acetate	61, 70, 88	$y = 13.0954x + 0.2275$	0.9834	74000	1100000 ±43700[a]	27	15	950000 ±86000[b]	3	13	269000 ±20400[c]	1	4
2	2-butanol	45, 59	$y = 3.7162x - 0.3587$	0.9959	660	3340±114[b]	27	5	4310±256[a]	27	7			
3	ethyl butanoate	71, 88	$y = 2.6487x - 0.0434$	0.9985	600	705000 ±23700[a]	3	1174	473000 ±5960[b]	3	789			
4	ethyl 2-methylbutanoate	57, 85, 102	$y = 2.7469 - 0.0477$	0.9980	74	2220±95.5[a]	9	30						
5	hexanal	56, 57, 72	$y = 1.5693x + 0.0533$	0.9995	80				1900±166[a]	3	24			
6	propyl butanoate	43, 71, 89	$y = 6.8926x - 0.0653$	0.9960	100	35900±3750[a]	9	358						
7	3-methylbutyl acetate	55, 70, 87	$y = 37.483x - 0.1104$	0.9975	200				2350±257[a]	81	12			
8	ethyl pentanoate	85, 88, 101	$y = 1.9027x - 0.0891$	0.9968	10	22300±904[b]	243	2234	25000±1400[a]	729	2497	4980±176[c]	27	498
9	ethyl 2-acetylbutanoate	73, 101, 116, 130	$y = 3.1531x - 0.152$	0.9922	-	39.2±6.75[a]	3	-				24.2±2.44[b]	3	-
10	methyl hexanoate	74, 87, 99	$y = 2.2227x - 0.0491$	0.9973	70	2480±329[a]	27	35	1790±167[b]	9	26			
11	ethyl hexanoate	88, 99, 101	$y = 1.5872x - 0.1239$	0.9977	76	3270000 ±112000[a]	19683	43057	3000000 ±200000[a]	19683	39450	1480000 ±14200[b]	2187	19413
12	ethyl (*E*)-3-hexenoate	69, 88, 142	$y = 12.672x - 0.1473$	0.9974	-	7.16±0.890[a]	3	-	2.96±0.450[b]	3	-			
13	ethyl (*Z*)-3-hexenoate	68, 69, 142	$y = 1.9997x - 0.0583$	0.9987	-				547±23.9[a]	9	-			
14	ethyl lactate	45, 75	$y = 1.515x - 0.0577$	0.9986	14000	7580±574[b]	3	< 1	14300±503[a]	3	1	5050±242[c]	3	< 1
15	isobutyl hexanoate	56, 99, 117	$y = 3.3022x + 0.0222$	0.9953	5250	6360±166[a]	3	1	5690±179[b]	3	1			
16	butyl hexanoate	56, 99, 117	$y = 9.5694x + 0.0397$	0.9990	700	5730±224[b]	729	8	7320±187[a]	729	10	433±39.7[c]	27	< 1
17	furfural	95, 96	$y = 15.011x - 0.0594$	0.9988	5800	72000±3280[a]	81	12	53800±1530[b]	9	9			
18	acetic acid	60	$y = 2.6083x - 0.0217$	0.9989	26000				24300±2080[a]	9	< 1			
19	tetramethylpyrazine	54, 136	$y = 1.6015x - 0.0192$	0.9968	690							75.7±4.80[b]	3	< 1
20	butyl lactate	56, 57	$y = 0.9677x - 0.0727$	0.9927	10000	2510±289[a]	9	< 1						
21	propyl octanoate	127, 145	$y = 1.0925x - 0.0324$	0.9908	-				814±77.4[a]	9	-			
22	octyl isobutanoate	71, 89, 112	$y = 2.0937x - 0.0839$	0.9976	6				31.5±2.94[a]	9	5			
23	nonyl acetate	56, 70, 83, 98	$y = 5.708x + 0.0051$	0.9814	600				765±24.6[a]	9	1			
24	phenylacetaldehyde	91, 92, 120	$y = 1.5714x - 0.0999$	0.9916	4	3250±165[b]	3	812	3760±229[a]	81	939			
25	1-nonanol	56, 70, 83, 98	$y = 15.454x + 0.447$	0.9864	34	210±24.0[a]	9	6	226±19.6[a]	81	7			
26	butanoic acid	60, 73	$y = 2.1023 + 0.0225$	0.9883	2500	86400±9130[a]	81	345	597000 ±4510[b]	729	239	273000 ±1070[c]	27	109

(*continued on next page*)

Table 2 (*continued*)

No.	Compound	Quantitive ions (*m/z*)	Linear equation	R^2	Threshold/ (μg/L)	C_{G1}/(μg/L)[a]	FD_{G1}[b]	OAV_{G1}[c]	C_{G2}/(μg/L)[d]	FD_{G2}[e]	OAV_{G2}[f]	C_{G3}/(μg/L)[g]	FD_{G3}[h]	OAV_{G3}[i]
27	diethyl butanedioate	101, 129	$y = 10.328x + 0.4551$	0.9762	75000	2370±125[b]	9	< 1				388±18.9[c]	3	< 1
28	2,6,6-trimethyl-2-cyclohexene-1,4-dione	68, 96, 152	$y = 1.9361x - 0.0219$	0.9940	25	65.7±8.00[b]	9	3						
29	6,10-dimethyl-2-undecanone	58, 71, 180	$y = 1.5976x + 0.0054$	0.9988	-				242±25.1[a]	3	-			
30	γ-hexalactone	70, 85	$y = 2.0386x - 0.0635$	0.9954	260							551±30.2[b]	27	2
31	phenyl acetaldehyde diethyl acetal	75, 103, 121, 149	$y = 4.1114x + 0.3506$	0.9816	-							262±26.2[c]	3	-
32	pentanoic acid	60, 73	$y = 0.9721x - 0.0223$	0.9987	500	68100±4300[a]	3	136						
33	5-ethyl-2(5*H*)-furanone	55, 83, 112	$y = 1.0163x + 0.084$	0.9904	-							64.5±6.87[a]	3	-
34	3-methylbutyl nonanoate	70, 71, 141, 159	$y = 2.5065x - 0.1843$	0.9926	-				17.0±1.25[a]	3	-			
35	methyl salicylate	92, 120, 121, 152	$y = 0.8338x - 0.0628$	0.9978	40				138±9.01[a]	9	3			
36	ethyl phenylacetate	91, 164	$y = 0.8522x - 0.08$	0.9981	73	1660±174[b]	9	23						
37	phenethyl acetate	91, 104, 105	$y = 2.2465x - 0.0132$	0.9941	108	231±16.5[b]	243	2	308±12.1[a]	729	3	11.4±0.280[c]	81	< 1
38	hexanoic acid	60, 73, 87	$y = 1.4479x - 0.0279$	0.9899	3000	1230000 ±33000[a]	729	409	914000 ±13900[b]	729	305	851000 ±26000[c]	27	283
39	ethyl 3-phenylpropanoate	91, 104, 178	$y = 2.7801x - 0.0097$	0.9949	-				1580±22.3[a]	3	-			
40	heptanoic acid	60, 73, 87	$y = 5.7291x + 0.0085$	0.9952	500	22300±389[a]	3	45				6470±112[c]	3	13
41	octyl octanoate	112, 127, 145	$y = 5.8652x - 0.1747$	0.9989	-				6090±205[a]	3	-			
42	γ-nonalactone	56, 85	$y = 3.5623x - 0.0985$	0.9936	21	210±19.1[a]	9	10	203±11.7[a]	3	10	32.5±2.51[b]	3	2
43	1-tridecanol	55, 69, 83, 97	$y = 2.3764x - 0.0384$	0.9978	-				6.81±0.970[a]	27	-			
44	*p*-cresol	77, 107, 108	$y = 1.8933x - 0.175$	0.9861	60	2270±223[a]	9	38						
45	phenethyl hexanoate	104, 105	$y = 3.3793x - 0.1502$	0.9924	156	111±13.6[b]	3	< 1				16.3±2.00[c]	3	< 1
46	methyl hexadecanoate	74, 87, 143	$y = 0.9723x + 0.1577$	0.9993	2000							54.3±4.91[b]	27	< 1
47	ethyl hexadecanoate	88, 101	$y = 4.0636x - 0.1674$	0.994	1000	4120±147.7[b]	9	4						
48	2,4-di-*tert*-butyl phenol	191, 192, 206	$y = 4.4355x + 0.0836$	0.9915	500	3730±207[a]	3	7				3190±265[b]	3	6
49	vanillin	151, 152	$y = 2.4795x - 0.0339$	0.9906	22	272±36.6[a]	6561	12						

The different letters marked with a, b and c on the compound concentrations represent significant differences between the data, Sig < 0.05, and there is no significant difference between the data marked with the same letters.

[a] Concentration in Grade 1 samples.
[b] FD factors in Grade 1 samples.
[c] OAVs in Grade 1 samples.
[d] Concentration in Grade 2 samples.
[e] FD factors in Grade 2 samples.
[f] OAVs in Grade 2 samples.

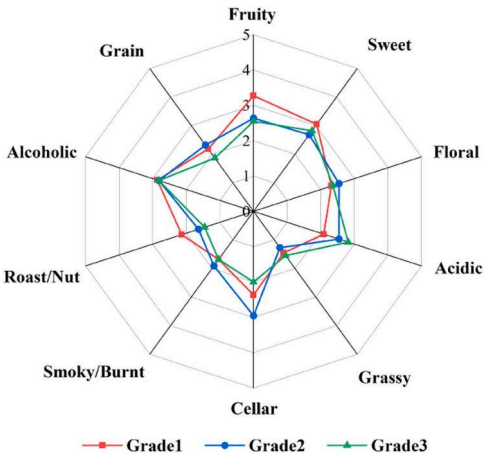


Fig. 1. Comparison of the three different grades of strong-aroma *Baijiu*.

in G1 sample, at 704 mg/L, followed by the G2 sample (473 mg/L), and the lowest content in G3 sample (203 mg/L). An appropriate amount of ethyl butanoate can impart a refreshing and clean taste to *Baijiu*. In addition, propyl butanoate, ethyl 3-hydroxyhexanoate, ethyl pentanoate, ethyl heptanoate, and ethyl octanoate also exhibited relatively high concentrations. Both the amounts and concentration of esters in G1 and G2 were much higher than those in G3 sample. These ester compounds imparted richer floral, fruity, and sweet aromas to G1 and G2 samples, making them more preferred by consumers.

A total of 17 acid compounds were identified in the three samples, and G1, G2, and G3 contained 14, 13, and 11, respectively. The highest acids concentration was observed in G1 sample at 2260 mg/L, followed by G2 sample at 1650 mg/L, and the lowest concentration was in G3 at 1160 mg/L. Among all the 17 carboxylic acids, hexanoic acid was the highest, accounting for more than half of the total acid contents (54.26 %, 55.50 %, and 73.25 % in G1, G2 and G3 respectively). The absolute concentration of hexanoic acid was the highest in the G1 sample and the lowest in the G3 sample, which was consistent with the order of ethyl hexanoate content. The content of hexanoic acid can affect the chemical balance of ethyl hexanoate, thus affecting *Baijiu*'s aroma characteristics (Hong et al., 2021). The content of butanoic acid ranked second among acid compounds; it usually has a fruity or cheesy aroma, which can contribute to *Baijiu* a refreshing feeling in taste (Wei et al., 2020). Hexanoic acid and butanoic acid can be biosynthesized by *Clostridium* using acetic acid or ethanol as substrates (Wu et al., 2023b; Yan and Dong, 2018). It was found that butanoic acid and hexanoic acid were the key aroma components of strong-aroma base *Baijiu*, through recombination and omission analysis (Dong et al., 2019). In all three samples, the contents of pentanoic acid and acetic acid ranked third and fourth, respectively, which contributed a sour aroma to the samples.

A total of 53 alcohol compounds were identified in the three samples; G1, G2 and G3 contained 31, 43, and 30, respectively. The mean concentrations of alcohol compounds in G1 and G2 samples were relatively high, at 150 and 128 mg/L, respectively. In contrast, the content in G3 sample was less than one-tenth of that in G1 and G2, at 12.5 mg/L. 1-Hexanol was the highest alcohol compound, which provided fruity and green aroma for *Baijiu*. Although its content was high, its high aroma threshold (5370 μg/L in 46 % ethanol aqueous solution) resulted in its low contribution to *Baijiu* aromas (Zhao et al., 2018). In G1 and G2 samples, the majority of alcohol compounds had six or fewer carbon atoms, whereas long-chain alcohols containing seven or more carbon atoms made up less than 10 %. However, in G3 sample, long-chain alcohols accounted for 22.82 % of the total content. Similar findings were also reported by Ding et al. (2015) and Zheng et al. (2014), who analyzed samples from different distillation stages. They observed that the content of low-molecular-weight alcohols was higher in the early stages of distillation, while the content of high-molecular-weight

[g] Concentration in Grade 3 samples.
[h] FD factors in Grade 3 samples.
[i] OAVs in Grade 3 samples.

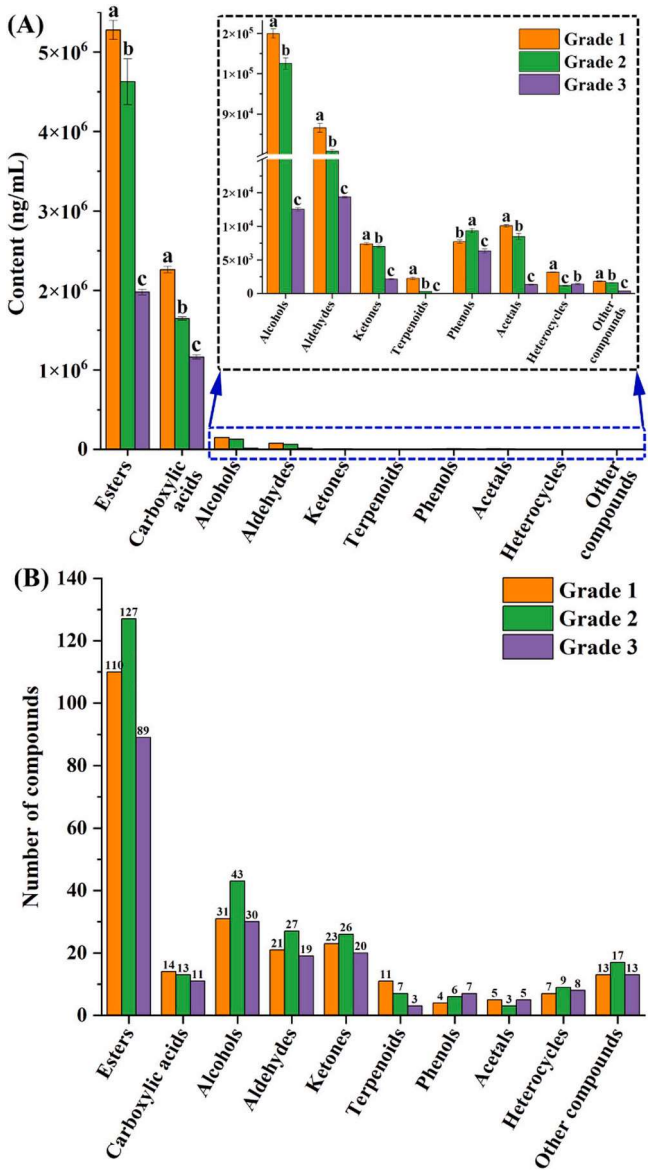


Fig. 2. The aroma compounds of the three grades of *Baijiu*. (A) The content of aroma compounds in three different grades of *Baijiu*, and the black dotted box is an enlarged view of alcohols, aldehydes, ketones, terpenes, phenols, acetals, heterocycles, and ether compounds. (B) The number of aroma compounds in three different grades of *Baijiu*.

alcohols was higher in the later stages. This is probably because the long-chain alcohols have higher boiling points, and the short-chain alcohols with lower boiling point volatilize in the early distillation process. Consequently, the content of long-chain alcohols in G3 sample was higher than that in G1 and G2 samples. This result was consistent with the findings of Li et al. (2012). In their study, it was observed that alcohols with six or fewer carbon atoms (such as 1-propanol, 1-butanol, isobutanol, etc.) mainly existed in the early fractions of distillation, while alcohols with higher boiling points, such as phenylethyl alcohol, primarily appeared in the later fractions of distillation. Research has shown that excessive levels of long-chain alcohols can lead to undesirable effects, such as hangovers (Tian et al., 2017). Hence, harmony in the variety and proportion of alcohols is crucial for both aroma and health.

A total of 31 aldehyde compounds were identified in three samples, with G1, G2, and G3 containing 21, 27, and 19 aldehydes respectively. The aldehyde contents in G1 and G2 were relatively high, at 79.8 and 62.2 mg/L respectively, while the content in G3 was much lower at 14.4 mg/L. Furfural with typical nutty, bready and roasted aroma, widely found in coffee (Chaichi et al., 2015), was detected in all three samples with concentrations of 72.0, 53.8, and 12.4 mg/L, respectively. It might impart a roast aroma to the *Baijiu* samples, and it was a key aroma compound of another premium strong-aroma *Baijiu*, *Wuliangye* (Niu et al., 2017). Furfural can be produced by microorganisms or generated through the Maillard reaction under high-temperature conditions during distillation (Zhang et al., 2019; Zhu et al., 2023).

A total of 33 ketone compounds were identified, including 23 in G1, 26 in G2 and 20 in G3. Their contents in G1 and G2 were 7.39 and 6.99 mg/L respectively, but in G3, the content was obviously lower than that in G1 and G2, only at 2.14 mg/L. Among these ketone compounds, 3-hydroxy-2-butanone (also named acetoin) had the highest contents in the three samples, at 2.75, 3.65, and 0.754 mg/L respectively, imparting a fruity aroma to *Baijiu*. It can be produced during fermentation by *B. subtilis* and can be derived into tetramethylpyrazine (Xu et al., 2018). In the study of Pu et al. (2021), it was found that the content of 3-hydroxy-2-butanone exhibited a noticeable upward trend with the increase of storage time in sauce-aroma *Baijiu*. The storage time of G1 and G2 was usually longer than that of G3, which may be one of the reasons for the relatively high content of 3-hydroxy-2-butanone in G1 and G2 samples. Ketone compounds mostly exhibited grassy, green, and fatty aromas, but due to their high aroma thresholds, their overall contribution was relatively low.

The three samples contained 11, 7 and 3 terpenes, respectively, and the terpene content in G1 (2.25 mg/L) was much higher than that in G2 (0.288 mg/L) and G3 (0.0175 mg/L). Their ratio was approximately 128:16:1. Terpenes are partly derived from raw grain materials and starter cultures, and partly from the conversion of other substances under high temperature in the distillation process (Feng et al., 2023). Wu et al. (2016) found that terpenes were important aroma-active compounds in sorghum. Similarly, Wang et al. (2016) found that terpenes could bring elegant and delicate aromas to *soy* sauce-aroma *Baijiu* according to olfactory results and OAVs. The content of longifolene among the terpenes was the highest in the three samples, at 1530, 156, 13.4 μg/L respectively, and their ratio was close to 114:12:1, accounting for more than half of the terpenes content (67.86 %, 54.20 %, 76.67 % in G1, G2 and G3 respectively).

Among the phenolic compounds, G2 sample had the highest content (9.34 mg/L), followed by G1 (7.71 mg/L), while G3 sample had the lowest content (6.33 mg/L). The content of 2,4-di-*tert*-butylphenol was the highest among the phenolic compounds in the three samples. 2,4-Di-*tert*-butylphenol is closely related to the burnt aroma of based *Baijiu*, which can increase the burnt aroma within a certain concentration range, and inhibit the burnt aroma beyond that range (Li et al., 2024). Some studies suggested that 2,4-di-*tert*-butylphenol originates from contaminants in the reagents used during the experiments, rather than from *Baijiu* itself. (Dong et al., 2023). The contents of 2,4-di-*tert*-butylphenol were followed by *p*-cresol and phenol, consistent with the quantitative results of Geng et al. (2023). The mixture of *p*-cresol and organic acids can produce a cellar-like aroma (Xiao et al., 2016). Phenol may be derived from the decomposition of tannins in the raw materials during the gelatinization process (Song et al., 2023).

Acetal compounds are usually regarded as the markers of *Baijiu* aging, and their content tends to increase during storage (Hong et al., 2021; Perestrelo et al., 2011; Thibaud et al., 2019). Acetals were formed from the condensation of aldehydes with alcohols (Fan and Qian, 2006; Zhu et al., 2016). Fan and Qian, (2006) found that 1,1-diethoxy-3-methylbutane played an important role in *Wuliangye* strong-aroma *Baijiu*, and its FD factor was $\geq$ 1024. However, its FD factors were 1 and 3 in G1 and G2 samples, respectively, and it was not detected in G3 sample. Four of the five acetals identified in this study were consistent with the identification results of Fan and Qian, (2006). The content of acetals in G3 (1.32 mg/L) was significantly lower than that in G1 (10.0 mg/L) and G2 (8.48 mg/L).

In the three samples, 7, 9 and 8 kinds of heterocyclic compounds were identified with contents of 3.17, 1.16, and 1.40 μg/L, respectively. These heterocyclic compounds were mainly pyrazines and furans, which provide roast, nutty and cocoa aromas to the samples. In contrast to the predominance of Maillard reaction products in baked goods, pyrazine compounds in *Baijiu* were primarily synthesized by microorganisms such as *Bacillus* species, with only a small amount formed *via* the Maillard reaction (Zhu et al., 2010). Zhang et al. (2019) confirmed that the contents of pyrazines increased during the distillation process due to the occurrence of Maillard reaction. Benzofurans and furanone compounds might have a positive contribution to the unique aroma. For example, Lee and Noble, (2003) confirmed that 2-propanoylfuran was associated with the oaky attributes of Californian Chardonnay wines.

3.3. Identification of aroma-active compounds in three different grades of Baijiu

AEDA is frequently used to assess the contribution of aroma compounds. In general, the higher the FD factor of an aroma compound, the more significant its contribution is deemed to be (Yang et al., 2022). Across the three samples, a total of 50 aroma-active compounds with FD factors ≥ 3 were identified, comprising 31 in G1, 31 in G2, and 18 in G3.

There were 30 ester compounds with FD factors ≥ 3, such as ethyl hexanoate, butyl hexanoate, and ethyl pentanoate. Ethyl hexanoate had the highest FD factor in all three samples, with values of 19683, 19683, and 2187, respectively. It is a key aroma compound in strong-aroma *Baijiu*, providing a fruity aroma to the samples (Mu et al., 2023; Song et al., 2020). Butyl hexanoate (FD = 729, 729, 27) and ethyl pentanoate (FD = 243, 729, 27) also had large FD factors and endowed samples with fruity aromas such as apple, pineapple, strawberry, etc. Phenethyl acetate (FD = 243, 729, 81) also imparted floral and honey aromas to samples. Overall, the FD factors of ester compounds in G1 samples were relatively large, which provided the samples with richer fruity aromas. This corresponded with the sensory evaluation findings of the three samples; the sensory scores for fruity attributes were 3.27, 2.64, and 2.55 for G1, G2, and G3, respectively. Among the ester compounds, the contents of ethyl esters accounted for 97.99 %, 98.14 % and 99.61 % in three samples, respectively, while propyl ester, butyl ester, pentyl ester, hexyl ester, heptyl ester, octyl ester, etc. only accounted for less than 3 % of the esters. However, these esters (such as butyl hexanoate) were aroma-active and contributed to the overall aroma of the samples.

Among the acidic compounds, there were five aroma-active compounds with FD factors ≥ 3. Hexanoic acid (FD = 729, 729, 27) and butanoic acid (FD = 81, 729, 27) had the highest FD factors, and both were produced by microbial fermentation and can react with ethanol to generate two major esters in *Baijiu*, ethyl hexanoate and ethyl butanoate (Yin et al., 2023). A low concentration of hexanoic acid can increase the fruity, floral, and sweet aromas of *Baijiu*, while a high concentration of butanoic acid can increase the floral aroma (Zhang et al., 2023). Acetic acid (FD = 1, 9, 1) and heptanoic acid (FD = 3, 1, 3) also exhibited aroma activity, providing *Baijiu* with a certain acidic, fruity, and cheese-like aroma.

The FD factors of most alcohol compounds were small, but there were 22 alcohols with aroma activity, such as 1-hexanol, 2,3-butanediol, 2-furanmethanol, etc. It is worth noting that 2-furanmethanol was only detected in G2 and G3, suggesting that it may be formed by Maillard reaction in the distillation process. Moreover, some low-molecular-weight alcohols (such as cyclopentanol, (*Z*)-2-penten-1-ol) were primarily detected in G1 and G2, while high molecular weight alcohols (such as 1-dodecanol) were mainly detected in G2 and G3; this difference could be due to their different boiling points.

There were four aldehyde compounds with FD factors ≥ 3, hexanal, furfural, phenylacetaldehyde, and vanillin. Hexanal has a typical grassy aroma, and its FD factors were 1 and 3 in samples G1 and G2, respectively, possibly providing a fresh aroma to the samples. Furfural may contribute sweetness and bread-like aroma to the *Baijiu* samples. The FD factors of phenylacetaldehyde in the three samples were 3, 81 and 1, respectively, possibly providing the *Baijiu* with honey and floral aromas. Vanillin was only detected in G1 sample with an FD factor of 6561, possibly providing floral, vanilla, and sweet notes to G1 sample.

Among the 33 ketone compounds, there were 13 aroma-active compounds, with only 3 having FD factors ≥ 3. Dihydro-5-pentyl-2(3*H*)-furanone (FD=9, 3, 3) is a compound with cocoa and butter aroma, which is one of the main aroma components of *Baijiu* and *Huangjiu* (Wang et al., 2019). It also functions as an indicator of beer aging, with its concentration gradually rising during the fermentation process (Tsuji and Mizuno, 2010).

2,4-Di-*tert*-butylphenol (FD=3, 1, 3) is an aroma-active compound with a relatively large FD factor among the phenols, and is one of the main aroma compounds of *Baijiu* (Pang et al., 2021). However, some reports suggest that this compound originates from contaminants in reagents rather than from *Baijiu* itself (Dong et al., 2023). The FD factors of *p*-cresol in G1 and G3 were 9 and 1 respectively, and it was not perceived by panelists in G2 samples. Some studies have demonstrated that *p*-cresol is closely related to the cellar aroma of strong-aroma *Baijiu* (Guo et al., 2022).

Among the heterocyclic compounds, only tetramethylpyrazine and 5-ethyl-2(5*H*)-furanone had FD factors of 3 in G3 sample, while they were not perceived in the other two samples. These two compounds imparted cocoa, nutty, and bread-like aromas to the samples (Huang et al., 2024). While other pyrazines and furanones were not perceived at the olfactory detection port, a variety of pyrazine or furanone compounds might contribute to the overall aroma of the samples due to synergistic effects (Niu et al., 2018).

3.4. Quantification of aroma-active compounds and OAVs

Usually, the contribution of the compound is measured by the ratio of its concentration to its threshold. The aroma compounds with FD factor ≥ 3 were quantified by external standard curve method, and the results were shown in Table 2. The OAVs were calculated based on the threshold values in ethanol aqueous solution.

In samples G1, G2, and G3, there were 25, 21, and 9 compounds respectively with OAVs ≥ 1. However, there were also several compounds whose OAVs could not be calculated due to the absence of their threshold, and the contribution of these compounds to the overall aroma should be verified in subsequent omission experiments. Among these aroma-active compounds, ethyl hexanoate had the highest OAVs, with values of 43057, 39450, and 19431, respectively, followed by ethyl pentanoate, with OAVs of 2234, 2497, and 498. Although ethyl acetate had a high content, its relatively high threshold (74000 μg/L) resulted in a smaller OAV (OAV=15, 13, 4), thus making its contribution to the overall aroma relatively low. On the whole, compounds with higher FD factors tended to have larger OAVs. However, some compounds, such as butyl lactate, diethyl butanedioate and phenethyl hexanoate, had higher FD factors, but their OAVs were less than 1. The main reason for this phenomenon may be the influence of the matrix. The measurement of FD factors was primarily influenced by their thresholds in air, while the calculation of OAVs was based on their thresholds in aqueous ethanol solution.

3.5. Principal component analysis (PCA)

All compounds detected by GC×GC–O–TOF MS were analyzed by PCA. The analysis revealed that the three different grades of *Baijiu* could be well classified into three categories, and they were distinctly separated from each other, indicating significant inter-group differences. The cumulative contribution rate of variance for the first two principal components was 97.8 %, indicating minimal information loss and largely preserving the original information. From the Biplot diagram (Fig. 3), it can be seen that the compounds contributing significantly to the classification of G1 include *δ*-cadinene, *α*-santalene, *α*-humulene,

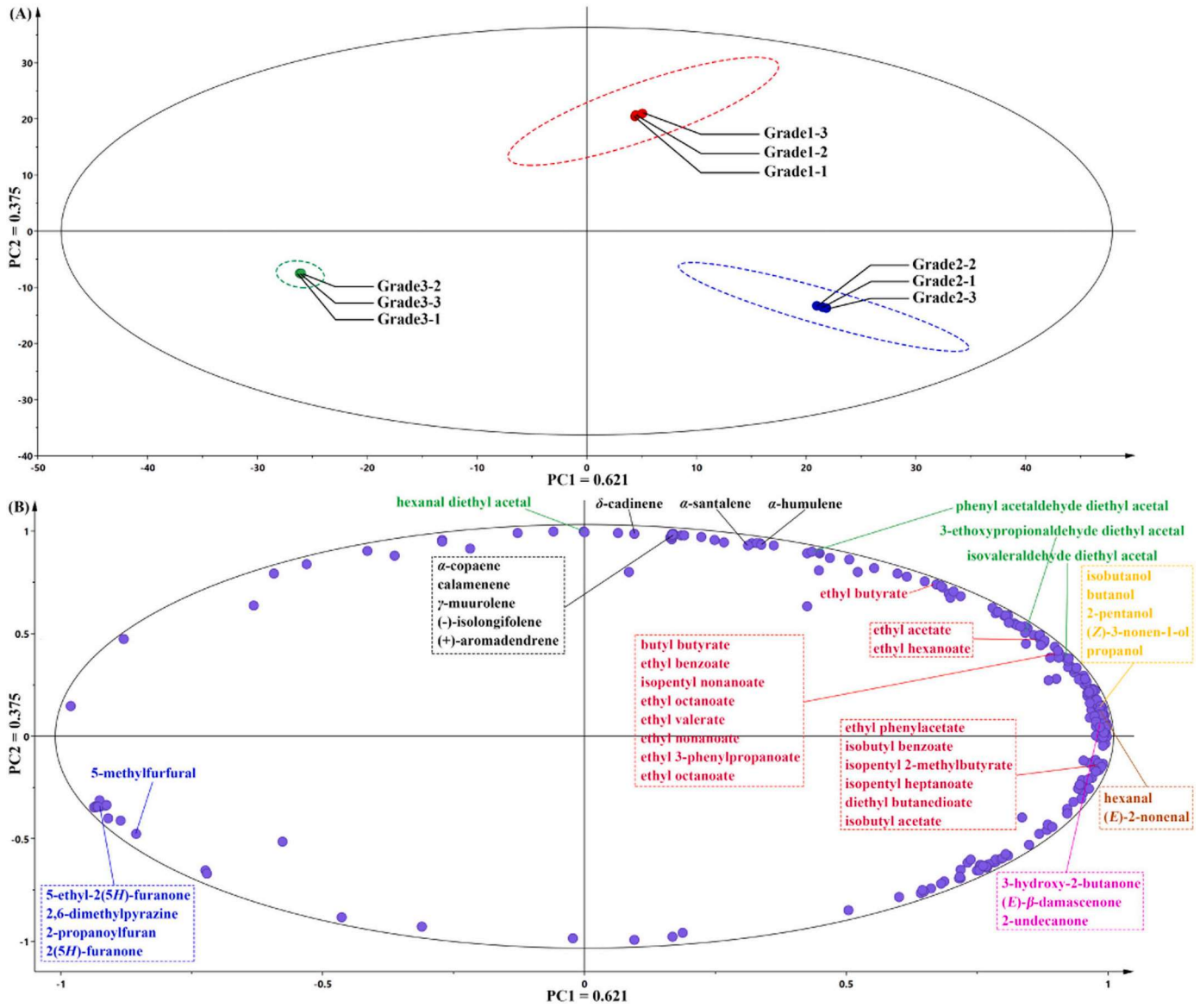


Fig. 3. Principal component analysis of the three different grades of *Baijiu*. (A) the score plot of the principal component analysis. (B) the loading plot of the principal component analysis.

α-copaene, calamenene, isolongifolene, γ-muurolene, aromadendrene, etc. In addition, acetal compounds such as phenylacetaldehyde diethyl acetal, hexanal diethyl acetal, and furfural diethyl acetal had strong positive correlations with G1 group, and they are generally considered as markers of *Baijiu* aging. The compounds that contribute to G2 group were mainly ester compounds, such as butyl hexanoate, ethyl dodecanoate, 3-methylbutyl lactate, etc., and these esters were also the major contributors of the first principal component. There were two main reasons for the difference between G3 and the other two grades. First, G3 sample contained fewer esters, alcohols, aldehydes, and ketones, with their contents being only less than half of that found in the other two grade samples. Second, G3 samples contained higher levels of compounds such as 5-methylfurfural, 2-propanoylfuran, 2,6-dimethylpyrazine, and 5-ethyl-2(5*H*)-furanone. It was speculated that these compounds were formed through the Maillard reaction under high-temperature conditions during distillation. These compounds imparted nutty and roasted aromas to G3 samples, but this could potentially lead to a deviation in their aroma profiles from the traditional styles of strong-aroma *Baijiu*.

3.6. Aroma recombination experiments

Through the sensory evaluation of 12 panelists (6 females and 6 males), a total of 10 aroma attributes were determined in the three samples, namely fruity, sweet, floral, acidic, grassy, cellar, smoky/burnt, roast/nut, alcoholic and grain. In order to verify the key aroma compounds in three different grades of *Baijiu*, aroma recombination experiments were conducted in simulated matrices. According to the ethanol content of the *Baijiu* samples, the simulated matrices of G1 and G2 were 52 % ethanol-in-water solutions, and the simulated matrix of G3 was a 42 % ethanol-in-water solution. The aroma compounds were added to the simulated matrices according to the content of external standard quantification method. It can be seen that, compared with the recombination model, most aroma attributes of the three different grades of samples were quite similar, with only a few aroma attributes showing some differences (Fig. 4). For example, in G1 sample and its recombination model, the five aroma attributes of alcoholic, smoky/burnt, cellar, grassy and sweet were relatively close, and the four aroma attributes in G1 were slightly higher than those of the recombinant model, while the floral aroma of the recombinant model was slightly higher than that

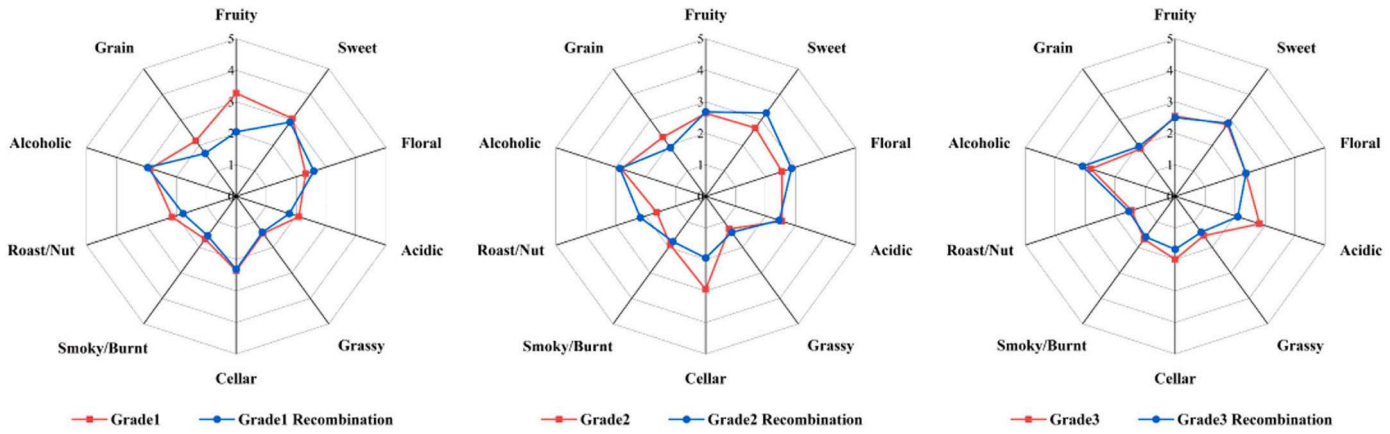


Fig. 4. Aroma characteristics of three different grades of *Baijiu* and their recombination models.

Table 3
Results of omission experiments for the three different grades of strong-aroma *Baijiu*.

No.	Compound	Grade 1[a]		Grade 2[b]		Grade 3[c]	
		N[d]	Significance[e]	***N***	Significance	***N***	Significance
1	ethyl acetate	9	**	9	**	8	*
2	2-butanol	4	-	4	-		
3	ethyl butanoate	10	***	9	**		
4	ethyl 2-methylbutanoate	4	-				
5	hexanal			8	*		
6	propyl butanoate	3	-				
7	3-methylbutyl acetate			4	-		
8	ethyl pentanoate	8	*	8	*	8	*
9	ethyl 2-acetylbutanoate	4	-			3	-
10	methyl hexanoate	2	-	4	-		
11	ethyl hexanoate	11	***	10	***	12	***
12	ethyl (*E*)-3-hexenoate	4	-	3	-		
13	ethyl (*Z*)-3-hexenoate			4	-		
14	ethyl lactate			9	**		
15	isobutyl hexanoate	5	-	4	-		
16	butyl hexanoate	4	-	4	-		
17	furfural	3	-	6	-		
18	propyl octanoate			4	-		
19	octyl isobutanoate			4	-		
20	nonyl acetate			4	-		
21	phenylacetaldehyde	8	*	8	*		
22	1-nonanol	8	*	5	-		
23	butanoic acid	7	-	8	*	5	-
24	2,6,6-trimethyl-2-cyclohexene-1,4-dione	3	-				
25	6,10-dimethyl-2-undecanone			3	-		
26	γ-hexalactone					4	-
27	phenylacetaldehyde diethyl acetal					5	-
28	pentanoic acid	7	-				
29	5-ethyl-2(5*H*)-furanone					3	-
30	3-methylbutyl nonanoate			5	-		
31	methyl salicylate			3	-		
32	ethyl phenylacetate	4	-				
33	phenethyl acetate	4	-	5	-		
34	hexanoic acid	10	***	9	**	9	**
35	ethyl 3-phenylpropanoate			4	-		
36	heptanoic acid	4	-			4	-
37	octyl octanoate			3	-		
38	γ-nonalactone	7	-	3	-	3	-
39	1-tridecanol			3	-		
40	*p*-cresol	4	-				
41	ethyl hexadecanoate	3	-				
42	2,4-di-*tert*-butyl phenol	7	-			3	-
43	vanillin	7	-				

[a] Grade 1 (*Tequ-Baijiu*)
[b] Grade 2 (*Tou-Baijiu*)
[c] Grade 3 (*Erqu-Baijiu*)
[d] Numbers of correct judgments in omission experiments from 12 assessors by the triangle test.
[e] Significance: *, significant ($p \leq 0.05$); **, highly significant ($p \leq 0.01$); ***, very highly significant ($p \leq 0.001$).

of G1. The five aroma attributes of G2 were close to its recombination model, such as alcoholic, smoky/burn, grassy, acidic and fruity, but the grain and cellar aroma of G2 was slightly higher than that of the recombination model, and the sweet, floral and roast/nut aroma of recombination model were higher than that of G2. The overall profiles of G3 and its recombinant model were quite similar, but the acidic and cellar aroma of G3 sample were slightly higher than that of the recombinant model. Through the recombination experiment on three samples, it was observed that the aroma of the two higher-grade samples, G1 and G2, were more complex, making it more challenging to simulate their profiles. Conversely, the aroma of the lower-grade G3 sample were relatively simpler, making its recombination easier.

3.7. Aroma omission experiments

To validate the contribution of each compound to the overall aroma, one compound was omitted from the recombination model each time, and then a triangle test was conducted to determine if the absence of that compound significantly affected the recombination model.

When ethyl hexanoate, ethyl butanoate, and hexanoic acid were omitted separately from the G1 recombination model, differences were perceived by 10 or more individuals compared to the original recombination model (Table 3), indicating that these three compounds had very important contributions to the aroma of G1 sample. When ethyl acetate was omitted from the G1 recombination model, differences were perceptible by 8 panelists, indicating that this compound had a significant impact on the model. When ethyl pentanoate, phenylacetaldehyde, and 1-nonanol were omitted from the recombination model in turn, sensory panelists could still perceive differences. However, when the other 20 compounds were omitted one by one, the differences in the model were not significant.

When ethyl hexanoate, ethyl acetate, ethyl butanoate, ethyl lactate, hexanoic acid, hexanal, ethyl pentanoate, phenylacetaldehyde and butanoic acid were omitted from G2 recombination model one by one, significant differences were perceived by the panelists between the omitted model and the recombination model. However, when the other 20 compounds, such as furfural, 1-nonanol, and 3-methylbutyl nonanoate were omitted individually, the differences were not significant. This implied that the first nine compounds were the key aroma compounds for G2 samples.

When ethyl hexanoate, hexanoic acid, ethyl acetate and ethyl pentanoate were omitted from the G3 recombination model sequentially, significant differences were observed between the omitted model and the recombination model. However, when eight compounds such as butanoic acid, phenylacetaldehyde diethyl acetal, γ-hexalactone, heptanoic acid, ethyl 2-acetylbutanoate, 5-ethyl-2(5*H*)-furanone, γ-nonalactone and 2,4-di-*tert*-butyl phenol were omitted in turn, the model difference was not significant. Therefore, ethyl hexanoate, hexanoic acid, ethyl acetate and ethyl pentanoate were determined to be the four key aroma compounds of G3 sample.

4. Conclusion

In this study, three different grades of strong-aroma *Baijiu* samples were systematically analyzed using molecular sensory science. Statistical analysis of the identified aroma compounds revealed that terpenes, acetals, and esters played significant roles in the high-quality strong-aroma *Baijiu*. Ester compounds were found to be the main contributors to mid-quality *Baijiu*. Two primary factors distinguished low-quality *Baijiu* from the former two. First, the low-quality *Baijiu* samples contained lower levels of ester compounds, about half of that in high-grade and mid-grade *Baijiu* samples. Secondly, the low-quality *Baijiu* samples contained compounds such as 5-methylfurfural, 2-acetylfuran, 2,6-dimethylpyrazine and 5-ethyl-2(5*H*)-furanone. These compounds were likely to be Maillard reaction products, generated under high-temperature conditions during the prolonged distillation process. The presence of these compounds could potentially alter the profile of the low-quality *Baijiu* samples, deviating from the distinctive characteristics typically associated with strong-aroma *Baijiu*. Most aroma attributes could be simulated through recombination experiments, though some differences remained in several attributes, which may be caused by interactions between aroma compounds or the influence of non-volatile compounds. Subsequently, the key aroma compounds of the three different grades of *Baijiu* were confirmed through aroma omission experiments. This study provided a basis for the classification of *Baijiu* quality, thereby promoting the high-quality development of the *Baijiu* industry.

CRediT authorship contribution statement

Baoguo Sun: Supervision, Funding acquisition. **Jinyuan Sun:** Supervision, Funding acquisition. **Fuping Zheng:** Supervision, Funding acquisition. **Qing Li:** Software. **Xiaojie Geng:** Software. **Zhiying Ma:** Investigation. **Fan Yang:** Writing – original draft, Formal analysis. **Erbao Chen:** Writing – original draft, Formal analysis.

Declaration of Competing Interest

There are no conflicts of interest to declare.

Data availability

No data was used for the research described in the article.

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